THE COSTS OF INFORMATION PROCESSING IN AN INFORMATION RATCHET SYSTEM

SUBMITTED
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Abstract

The Landauer's Principle states that the erasure of a bit of information must dissipates $k_B \ln 2$ amount of entropy. This spawned the field of information thermodynamics, studying how information are processed, and how it relates to the entropy of a thermodynamic system. However, Landauer's Principle fail to account for the structural costs associated with the implementation of the information processing devices, resulting in inefficient information processing. By studying the information ratchet system, which mediates interactions between a heat, work and information reservoirs, the relationship between information changes and energy flows can be quantified. A program is written to simulate the information ratchet system, allowing us to identify the need for a device to synchronise to its environment as a condition for efficient information processing. The existence of an unsurpassable bound that limits efficiency is also found. By studying the transient and modularity dissipations of information processing, it is clear that these device structural costs comes from the dissipation of unaccounted correlations between the device and the information reservoir, and the attempt to minimise these leads to the need for a device to synchronise to its environment.

Chapter 1

Introduction

Information are all around us in our modern digital age. They are constantly being created, stored, shared, and erased since the advent of digital technologies. One might not think much when we pick up a newspaper, or when we consume information in general, in fact one might see information as a free resource that comes without costs.

However, modern physics of information tells us that there is a cost associated with information, that other than the creation and storage of information, even the act of erasing information requires energy to be expended. This is the famous Landauer's Principle [1], which quantified the costs associated with the burden of information. Together with the development of information theory by Claude Shannon, information is seen as a physical quantity, and no longer an abstract idea. To appreciate these ideas, we need to have a short historical review.

1.1 Background

It has been more than a hundred years since James Clark Maxwell proposed his 'intelligent being' in *Theory of Heat*, his 1871 book [2]. This 'intelligent being', later referred to as Maxwell's Demon, lies at the centre of a box containing air at uniform temperature, partitioned into side A and B with a small hole at the middle. This demon, being intelligent, and with heightened senses is able to perceive the molecules, and without any input of work, can open and close the hole on the partition allowing faster molecules to pass from side A to B, while letting slower molecules to pass from side B to A. This raises side B's temperature, while lowering side A's, violating the Second Law of Thermodynamics, as this temperature difference can be used to do work.

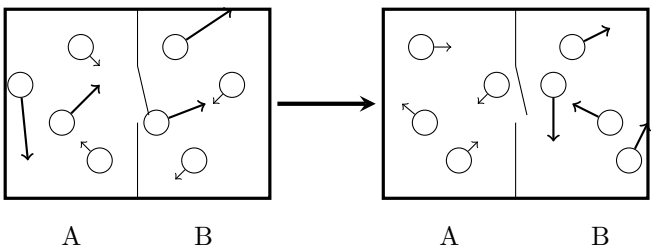


Figure 1.1: A box partitioned into side A and side B. Maxwell's Demon sort the box to fast molecules on side B and slow molecules on side A. This type of demon that create a temperature difference is called a temperature demon.

Although initially it seems like this is a consequence of conscious or intelligent intervention in a thermodynamic system [3], physicists were quick to design mechanical demons that replace the need for a conscious intelligent being. One of such mechanism was a trapdoor held closed by a spring, conceived by Smoluchowski in 1912 [4]. The trapdoor is to be designed such that it can only be open in a single direction. Whenever a molecule hit the trapdoor from side A, it will be swung open, allowing the molecule

to pass into side B. It will immediately closes itself as the spring pushes the trapdoor back. Whereas if molecules from side B were to hit the trapdoor, it cannot be opened. This is a pressure demon which creates a density or pressure difference which can be used to do work.

The trapdoor design would not work as it will become thermalised with the system, and will exhibit thermal motions as it swing open and close even without being hit by a molecule. Computational simulations have been done in recent years to demonstrate that the trapdoor indeed resulted in no net flow [5], [6]. Although Smoluchowski's trapdoor fails to function as a Maxwell's Demon, it has directed the solution from one of intelligent intervention to one of engineering, that perhaps it is physically impossible to build a Maxwell's Demon.

In 1929, Leo Szilard proposed his ground-breaking one-molecule engine [7] as shown in Figure 1.2. A partition is inserted into a box containing a single molecule with two pistons at the side. At this point, the molecule is either in the left or the right partition. A measurement is made to determine which partition the molecule is in. If the molecule is in the right partition, the left partition's piston is compressed with no work done as we are compressing vacuum. The partition is then removed and the molecule will push the left partition back, doing work in the process and absorbing the lost energy as heat from the environment.

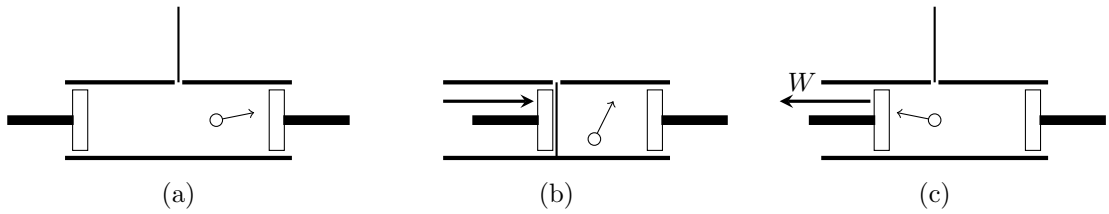


Figure 1.2: The Szilard Engine. (a) A single molecule is in a box with two pistons at the side. (b) A partition is inserted at the centre of the box, the piston at the side without the molecule is compressed. (c) The partition is removed, and the molecule pushes the piston back, this can be use to do work—e.g. lifting a weight. The molecule absorb heat from the environment to make up for the lost of energy.

The Szilard engine is able to produce work in a thermodynamic cycle by absorbing surrounding heat, violating the Kelvin-Planck statement of the Second Law. However, Szilard directs the attention to the act of measurement. The engine will only work if we know which partition the molecule is in. This lay the groundwork for the relationships between measurement, information, and entropy.

During this period of time, Quantum Mechanics were established and the costs of measurement were associated with the measurement problem in Quantum Mechanics as well as the uncertainty principle. In 1951, Leon Brillouin investigated the act of measurements in Maxwell's Demon devices [8], [9]. To determine the position or momentum of a particle, we are required to scatter at least one photon off it, and Brillouin demonstrated that a minimum entropy of $k_B \ln 2$ per particle were produced in this act of measurement, saving the Second Law in the Szilard engine.

Just a few years prior, *The Mathematical Theory of Communication* was formulated by Claude Shannon and Warren Weaver [10], where the connections between information and entropy were established. Shannon's information entropy and the thermodynamic entropy will eventually be united [11], and information will be forced to be treated as a physical quantity.

In 1961, Rolf Landauer suggested that the focus should be on information itself, rather than the act of gathering information or measurement [1]. Maxwell's Demon engine must be a cyclic process, such that the state of the system at the end must be the same as the start. Treating information as physical, this require the erasure of the previously stored information—e.g. knowing which partition the molecule is in. This irreversible erasure of information was compared to the irreversibility of heat flow in the Second Law. 'Information is physical', a famous quote of Landauer, is now widely accepted in modern physics.

Together with Charles Bennett, whom also made major contributions to the field of quantum information [12], Landauer and Bennett were able to design a modified Szilard engine that removes the measuring process, demonstrating once and for all that entropy production comes from information erasure [13]–[16]. This will come to be known as the Landauer's Principle, whereby the erasure of 1 bit of information generates $k_B \ln 2$ amount of entropy. This generation of entropy saves the Second Law as it offsets the entropy reduction by Maxwell's Demon.

1.2 Information Processing and the Second Law

Today, the study of the Second Law of Thermodynamics and the physics of information has given us two main results. First, there is the fluctuation theorem, which states that the Second Law is only statistically true—negative entropy production is possible, but are exponentially rare [17]–[19]:

$$\frac{\Pr(\Delta S)}{\Pr(-\Delta S)} = e^{\Delta S} \tag{1.1}$$

$$\implies \langle \Delta S \rangle \geq 0 \tag{1.2}$$

Whereby ΔS is the change in entropy, and $\Pr(X)$ denotes the probability of X. Second, a modified Second Law was proposed that accounts for the informational entropy of a thermodynamic system [20]:

$$\langle \Delta S \rangle + k_B \ln 2 \; \Delta H \geq 0 \tag{1.3}$$

Whereby ΔH is the change in Shannon information entropy. This modified Second Law is an extension of the thermodynamic fluctuation theorem in equation 1.2, such that if $\langle \Delta S \rangle$ were to decrease, an increase in the Shannon information entropy must be seen. The Landauer's Principle is also apparent in this modified Second Law—an erasure of 1 bit correspond to $\Delta H = -1$, guaranteeing an increase in entropy of at least $k_B \ln 2$:

$$\langle \Delta S \rangle \geq k_B \ln 2 \tag{1.4}$$

The modified Second Law put information entropy on the same footing as the thermodynamic entropy in a thermodynamic system. This sparks the need for a Physical Theory of Information, whereby information dynamics are given the same treatment as energy dynamics. As chemical, kinetic, thermal, and gravitational phenomena were generalised by energy dynamics, the underlying principle behind uncertainty, communication, causality, and entropy will be described by information dynamics.

Modern studies of information dynamics generally lies in three categories: (1) Measurements and feedback controls, similar to the original Szilard engine [21], [22]. (2) Interactions and information exchanges between multipartite systems whereby information shared by systems known as mutual information, and entropy dissipation from information transfer known as transfer entropy are of concern [23], [24]. (3) Physical thermodynamic system with the inclusion of an information reservoir, besides the usual heat and work reservoir [25]. All three categories, through the modified Second Law, study the costs of the creation of information, its transfer, how it changes from one form to another, how it is being shared between systems, and the erasure of it. These are known as information processing, as energy thermodynamics processes energy, information thermodynamics processes information.

1.3 The Costs of Information Processing

The focus of our study lies in the third category, of the inclusion of an information reservoir in a physical thermodynamic system. One effective and widely used model is the information ratchet [26], [27], which

uses a tape of string as its information reservoir. The information ratchet is useful as it is a device that directly mediates interactions between the heat, work and information reservoirs. The creation or erasure of information in the information reservoir is coupled to the energy flows between the device and the heat and work reservoirs. Hence, the dynamics between the Maxwellian work produced by Maxwell's Demon functionalities and the amount of information entropy generated can be quantified.

The information ratchet has real world parallels like DNA polymerase on nucleotide base-pairs [28], and designs of autonomous biomolecules or nano-machines that functions by Maxwellian work were also proposed [29]. However, the ratchet device is also a variable in the system. A ratchet might hoard some amount of information or energy, affecting the achievability of the equality in the modified Second Law, decreasing the efficiency of the ratchet at information processing. The study of ratchet efficiencies have been the focus of recent studies [27], [30], [31].

These losses in efficiencies are due to various dissipations in the system. As information are transferred, shared or lost within the system, some amount of dissipations are present. Different ratchet devices processes information differently, some ratchets experience dissipations more than others and are thus less efficient. This is referred to as the structural costs of the device's implementation, and they directly affects the efficiency of the device's performance in transducing information, or in producing Maxwellian work.

Is there an intrinsic structural cost to information processing devices that is not captured by Landauer's Principle, and if there is, is there a bound that cannot be surpassed? Is there a best structural design for a given information content, and if there is, what are the conditions for the design? These are the questions that we hope to answer in this report.

We hope to answer these questions by comparing ratchets of different structures. This is done by writing a program that allow us to easily simulate different ratchet designs, as well as information reservoirs of different information contents. The efficiencies or performance of different ratchet designs will be compared with each other, enabling us to identify certain structural conditions required for efficient information processing.

However, to find the origins of these efficient structural conditions, we need to look at some known costs or dissipations of information processing. The information-theoretic conditions to minimise these costs and dissipations can be matched with the efficient structural conditions of our ratchet, allowing us to determine the type of costs and dissipations that are associated with it

The physics gained from studying the efficiencies of the mediating devices for information processing are important to the field of the physics of information, as well as allowing better designs of computing devices, or to better understand the functioning of biological enzymes that leverage environmental information to function like a Maxwell's Demon.

1.4 Chapters Review

Our study of the costs of information processing in the information ratchet system require us to first define the information ratchet system formally in Chapter 2. In Chapter 2, we will describe how the system transiting from one state to another is governed by Markov processes. This is followed by a discussion on the energy dynamics of the ratchet, and how it exchanges heat and work, which is summarised by Boltzmann statistics. A new Second Law will be introduced, and we will define how information is measured in the information reservoir and the ratchet. Lastly, we will give a simple demonstration of our simulation program written in *Python*, and compare the numerical results with the analytical results.

In Chapter 3, we will begin investigating different ratchet system using our program. The new Second Law will be demonstrated to be a better bound compared to the modified Second Law in equation 1.3. We will then demonstrate that the need for a ratchet to synchronise is an important condition for efficient information processing, by showing how a self-synchronising ratchet have better functionalities. Our ratchet design will then be compared to another existing ratchet design, and we will demonstrate that they have almost the same functionalities, which hints at a principle for efficient information pro-

cessing known as the principle of Requisite Complexity. Lastly, we will demonstrate that there is a bound that cannot be surpassed even with the most efficient ratchet design.

In Chapter 4, we will investigate two dissipations in information processing, known as the transient dissipation and the modularity dissipation. These are the structural costs associated with the adaptation of a system to its environment, and of doing information processing modularly. We will demonstrate which dissipation the conditions and principles that we identified in Chapter 3 correspond to. These dissipations will also be minimised to determine the most efficient type of ratchets. Lastly in Chapter 5, we will conclude the report by summarising the results and provide some insights on future works.

Chapter 2

The Information Ratchet

2.1 Introduction

The information ratchet is an abstract physical model of a ratchet moving unidirectionally along a tape of binary strings as shown in Figure 2.1. The system consists of 4 parts:

1. An infinite tape of symboled strings serving as the information reservoir.
2. A mass depicted as being lifted by a pulley as the work reservoir.
3. A heat reservoir at a temperature T.
4. A finite-state ratchet that governs and mediates interactions between all three reservoirs.

The ratchet travels unidirectionally along the tape, exchanging energies with the heat and work reservoirs, while writing or erasing information on the tape by changing the individual symbols the ratchet is on. The ratchet will extract heat from the heat reservoir, converting it to work, seemingly violating the Kelvin-Planck statement of the Second Law of Thermodynamics. This Maxwell's Demon engine operation is only possible with the inclusion of the information reservoir, whereby the entropic costs required by the Second Law are offset by the change in entropy of the information reservoir.

Though abstract, the information ratchet should remind us of a Turing machine, moving its reader along the tape while changing the tape's symbols and its own internal states, doing computations in the process. The ratchet also has biological parallels such as a DNA polymerase scanning and copying nucleotides, or other types of biomolecules and enzymes functioning that has the same behaviours as a protein motor [28], [32]. These hints at multiple possible future applications that comes with the study of the information ratchet.

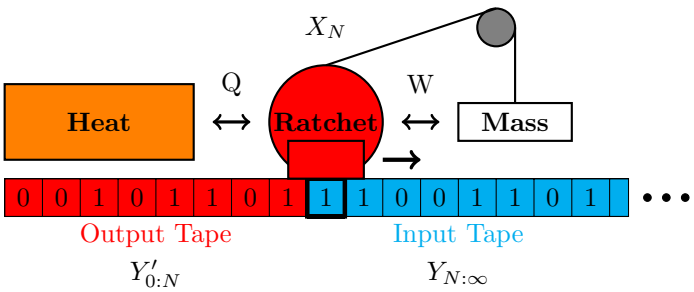


Figure 2.1: Information ratchet on the N_{th} symbol with ratchet state X_N, exchanging heat Q and work W, while moving along the input tape $Y_{N:\infty}$ and writing new symbols to the output tape $Y'_{0:N}$.

Physically, the idea of a ratchet extracting heat and storing it as work comes from the ratchet and pawl model of the Brownian or Feynman-Smoluchowski ratchet [33], [34]. Whereby random thermal

fluctuations bombards a paddle connected to a ratchet, which can only be rotated in one direction due to a pawl interlocking with the ratchet. The ratchet's rotation can then be used to raise a weight to do work, similar to the pulley in Figure 2.1. The Brownian ratchet does not work in practice due to thermal motions appear in the pawl when it became thermalised, rendering the interlocking mechanism useless. However, we shall assume that our information ratchet extract heat from the heat reservoir storing it as work in the raised weight in a similar way to the Brownian ratchet. With the information reservoir, our information ratchet is capable of functioning as a Maxwell's Demon in certain non-equilibrium cases, unlike the Brownian ratchet.

As for writing symbols to the tape, we can assume similar workings to a Turing machine, having a reader that reads and writes symbols to the tape. While there are multiple information-bearing degrees of freedom in actual physical systems, this model only have a single degree of freedom of the tape. This will suffice for our study, noting that the information ratchet is but an abstract model for us to extract the relevant physics of information processing.

2.2 States and Transitions

We will now formally define the workings of the information ratchet. The finite-state ratchet has a finite number of internal states $\mathcal{X}$, similar to the internal states of a Turing machine, while the symboled tape have symbols $\mathcal{Y} \in \{0, 1\}$. At each interval, the ratchet is on a single symbol on the tape, resulting in a joint ratchet-tape states of $\mathcal{X} \otimes \mathcal{Y}$. For example, a ratchet with two internal states A and B will be referred to as a 2-states ratchet, and will have:

$$\begin{aligned} \mathcal{X} &\in \{A, B\} \\ \mathcal{Y} &\in \{0, 1\} \\ \mathcal{X} \otimes \mathcal{Y} &\in \{A \otimes 0,\ A \otimes 1,\ B \otimes 0,\ B \otimes 1\} \end{aligned}$$

For convenience, we will omit the joint product symbol '$\otimes$' for the joint states from this point on, such that $A \otimes 0$ will be written as $A0$ and similarly for other joint states. Note that the calligraphic labels $\mathcal{X}$, $\mathcal{Y}$, and $\mathcal{X} \otimes \mathcal{Y}$ denotes the symbols or alphabets of the actual random variables X_N, Y_N, and $X_N \otimes Y_N$, such that the actual states are drawn from the calligraphic sets, with N labelling the N_{th} symbol on the tape. For example, in the case of the 2-states ratchet, the possible values of X_N are $X_N = A$ and $X_N = B$ with $\mathcal{X} \in \{A, B\}$. We will use a lower case letter when referring to individual symbols in the calligraphic sets. Therefore, $X_N = x_N$ with $x_N = A$ or $x_N = B$.

The ratchet will start at the very first symbol of the tape Y_0 with an initial ratchet state X_0, having a joint state of $X_0 \otimes Y_0$. The ratchet will then interacts with the heat reservoir for an interval τ, and transiting to the joint state $X_1 \otimes Y_0'$. This means that the ratchet state transits to X_1 and changed Y_0 to Y_0' at time $t = \tau$. The ratchet will then interacts with the work reservoir and move on to the next symbol Y_1, forming the new joint state of $X_1 \otimes Y_1$, and the same process continues.

For example, at $t = N\tau$, if the ratchet state is at $X_N = A$ and the incoming symbol is $Y_N = 1$,

$$\begin{aligned} \text{Joint state at } t = N\tau: &\quad X_N \otimes Y_N = A1 \\ \text{Interaction/Transition}\ : &\quad X_N \otimes Y_N \to X_{N+1} \otimes Y_N' = A1 \to B0 \\ \\ \therefore X_{N+1} = B, &\quad Y_N' = 0 \\ \text{If next incoming symbol}\ : &\quad Y_{N+1} = 1 \\ \text{Joint state at } t = (N+1)\tau: &\quad X_{N+1} \otimes Y_{N+1} = B1 \\ &\quad \vdots \end{aligned}$$

Going through this process from Y_0 until Y_N at $t = N\tau$ will output a new sequence of symbols $Y_{0:N}'$

(shown in red in Figure 2.1), whereas the yet to be processed sequence is $Y_{N:\infty}$ (shown in cyan). The transition from $X_N \otimes Y_N \to X_{N+1} \otimes Y'_N$ is governed by a Markov transition matrix M:

$$M\vec{\rho}(N) = \vec{\rho'}(N) \tag{2.1}$$

Where $\vec{\rho}(N)$ and $\vec{\rho'}(N)$ are the probability state vectors of the joint states before and after transition respectively, with $\Pr(a = b)$ denoting the probability of $a = b$.

$$\vec{\rho}(N) = \begin{pmatrix} \Pr(X_N \otimes Y_N = A0) \\ \Pr(X_N \otimes Y_N = B0) \\ \Pr(X_N \otimes Y_N = C0) \\ \vdots \\ \Pr(X_N \otimes Y_N = A1) \\ \Pr(X_N \otimes Y_N = B1) \\ \Pr(X_N \otimes Y_N = C1) \\ \vdots \end{pmatrix}, \quad \vec{\rho'}(N) = \begin{pmatrix} \Pr(X_{N+1} \otimes Y'_N = A0) \\ \Pr(X_{N+1} \otimes Y'_N = B0) \\ \Pr(X_{N+1} \otimes Y'_N = C0) \\ \vdots \\ \Pr(X_{N+1} \otimes Y'_N = A1) \\ \Pr(X_{N+1} \otimes Y'_N = B1) \\ \Pr(X_{N+1} \otimes Y'_N = C1) \\ \vdots \end{pmatrix} \tag{2.2}$$

$\vec{\rho}(N)$ and $\vec{\rho'}(N)$ contains all the joint states of the ratchet-tape system ordered with the '0' symbols first before '1'. Their respective elements sum up to unity. The transition matrix M is defined as:

$$M = \begin{pmatrix} M_{A0\to A0} & M_{B0\to A0} & \cdots & M_{A1\to A0} & M_{B1\to A0} & \cdots \\ M_{A0\to B0} & M_{B0\to B0} & \cdots & M_{A1\to B0} & M_{B1\to B0} & \cdots \\ M_{A0\to C0} & M_{B0\to C0} & \cdots & M_{A1\to C0} & M_{B1\to C0} & \cdots \\ \vdots & \vdots & \ddots & \vdots & \vdots & \cdots \\ M_{A0\to A1} & M_{B0\to A1} & \cdots & M_{A1\to A1} & M_{B1\to A1} & \cdots \\ M_{A0\to B1} & M_{B0\to B1} & \cdots & M_{A1\to B1} & M_{B1\to B1} & \cdots \\ M_{A0\to C1} & M_{B0\to C1} & \cdots & M_{A1\to C1} & M_{B1\to C1} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \tag{2.3}$$

With $M_{x_N \otimes y_N \to x_{N+1} \otimes y'_N} = \Pr(X_{N+1} = x_{N+1}, Y'_N = y'_N \mid X_N = x_N, Y_N = y_N)$, and we have written as such to make the matrix compact. This means that each element of the matrix denotes the probability of transition from one state to another. For example, $M_{B0\to A1}$ is the probability of transitioning the joint state from $B0$ to $A1$. Note that the transition matrix M is a left-handed matrix, such that its column elements sum up to unity instead of the row elements. M must also be a square matrix, with its number of columns being the length of $\mathcal{X}$ multiplies by the length of $\mathcal{Y}$.

As an example, we will define an arbitrary transition matrix M for the 2-states ratchet to describe the transition of $X_N \otimes Y_N = A1$:

$$M\vec{\rho}(N) = \vec{\rho'}(N)$$

$$\begin{pmatrix} M_{A0\to A0} & M_{B0\to A0} & M_{A1\to A0} & M_{B1\to A0} \\ M_{A0\to B0} & M_{B0\to B0} & M_{A1\to B0} & M_{B1\to B0} \\ M_{A0\to A1} & M_{B0\to A1} & M_{A1\to A1} & M_{B1\to A1} \\ M_{A0\to B1} & M_{B0\to B1} & M_{A1\to B1} & M_{B1\to B1} \end{pmatrix} \begin{pmatrix} \Pr(X_N \otimes Y_N = A0) \\ \Pr(X_N \otimes Y_N = B0) \\ \Pr(X_N \otimes Y_N = A1) \\ \Pr(X_N \otimes Y_N = B1) \end{pmatrix} = \begin{pmatrix} \Pr(X_{N+1} \otimes Y'_N = A0) \\ \Pr(X_{N+1} \otimes Y'_N = B0) \\ \Pr(X_{N+1} \otimes Y'_N = A1) \\ \Pr(X_{N+1} \otimes Y'_N = B1) \end{pmatrix}$$

$$\begin{pmatrix} 0 & 0.3 & 0 & 0 \\ 1 & 0 & 0.6 & 0 \\ 0 & 0.7 & 0 & 1 \\ 0 & 0 & 0.4 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0.6 \\ 0 \\ 0.4 \end{pmatrix}$$

Therefore, $\Pr(X_{N+1} \otimes Y'_N = B0) = 0.6$ and $\Pr(X_{N+1} \otimes Y'_N = B1) = 0.4$. This arbitrary Markov transition matrix M will have the Markov chain shown in Figure 2.2. It is important to note that this Markov chain is for a single interval τ of the joint states transition, rather than the Markov chain of the overall joint states transitions throughout the entire tape.

To illustrate the difference, consider $X_N \otimes Y_N = A0$, which transits to $X_{N+1} \otimes Y'_N = B0$ according

to Figure 2.2. If the next incoming tape symbol is $Y_{N+1} = 1$, the next joint state is $X_{N+1} \otimes Y_{N+1} = B1$ which then transits to $X_{N+2} \otimes Y'_{N+1} = A1$ with probability 1. The overall joint states transition would then have the transition

$$X_N \otimes Y_N \to X_{N+1} \otimes Y'_N \to X_{N+1} \otimes Y_{N+1} \to X_{N+2} \otimes Y'_{N+1} = A0 \to B0 \to B1 \to A1$$

However, the transition of $B0 \to B1$ is forbidden in the Markov chain in Figure 2.2. As such, to make our discussion clearer, it is convenient to classify transitions into four categories:

1. Input joint states transitions: $X_N \otimes Y_N \to X_{N+1} \otimes Y_{N+1} \to X_{N+2} \otimes Y_{N+2} \ldots$
2. Output joint states transitions: $X_{N+1} \otimes Y'_N \to X_{N+2} \otimes Y'_{N+1} \to X_{N+3} \otimes Y'_{N+2} \ldots$
3. Interaction joint states transitions: $X_N \otimes Y_N \to X_{N+1} \otimes Y'_N$
4. Switching joint states transitions: $X_{N+1} \otimes Y'_N \to X_{N+1} \otimes Y_{N+1}$

Thus, the $B0 \to B1$ transition is actually a switching transition which is not described by transition matrix M, and the overall joint state transition is made up of alternating interaction and switching transitions. Figure 2.3 summarises the overall joint states transitions. One way that we can make the transitions clearer is to bracket the interaction joint state transitions. Such that intra-brackets transitions are interaction transitions and inter-brackets transitions are switching transitions:

$$(X_N \otimes Y_N \to X_{N+1} \otimes Y'_N) \to (X_{N+1} \otimes Y_{N+1} \to X_{N+2} \otimes Y'_{N+1}) = (A0 \to B0) \to (B1 \to A1)$$

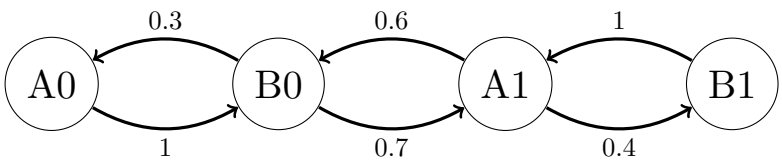


Figure 2.2: Markov chain of interaction joint states transition $X_N \otimes Y_N \to X_{N+1} \otimes Y'_N$ for arbitrary transition matrix M of a 2-states ratchet, with ratchet states A and B, and tape symbols 0 and 1.

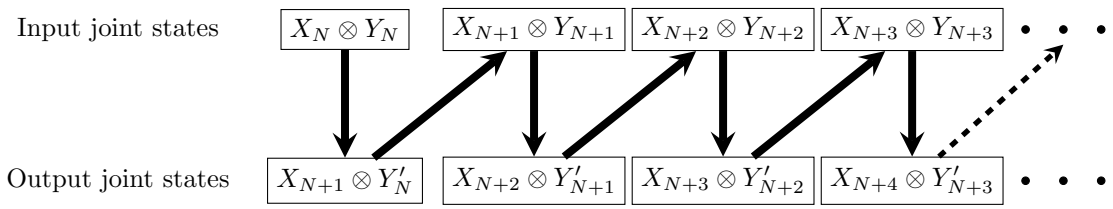


Figure 2.3: The arrows show the overall joint states transitions. The vertical downward arrows show the interaction transitions, whereas the diagonal upward arrows show the switching transitions. All the joint states before the interaction transitions are the input joint states, and all the joint states after are the output joint states. The dashed arrow shows the transition to joint states that are not shown in the figure.

One final point to note is that the choices of $\mathcal{X} \in \{A, B, C, \ldots\}$ and $\mathcal{Y} \in \{0, 1\}$ are arbitrary. The ratchet states can be for example $\mathcal{X} \in \{S_0, S_1, S_2, \ldots\}$, and the tape symbols $\mathcal{Y}$ are chosen as such in relation to the binary bits in computing, but can has more than two symbols such as $\mathcal{Y} \in \{A, T, G, C\}$ for DNA bases. In an effort to be general, we have referred to the symbols of $\mathcal{Y}$ as 'symbols' in this section. In all later sections, we will use 'bits' interchangeably with 'symbols' as we take $\mathcal{Y} \in \{0, 1\}$.

2.3 Energetics and Ratchet Dynamics

Different from a Turing machine, and in favour of biological parallels, each ratchet states and bits on the tape can has differing energies. This is similar to biochemical systems whereby some bonds are more favourable than others.

Imagine if enzyme $X_N = A$ prefers to bond with substrate $Y_N = 0$ more than substrate $Y_N = 1$, we would then expect energy $E_{A0} < E_{A1}$. Additionally, if $E_{B0} < E_{A1}$, then there is a higher probability for $A0 \to B0$ than for $A0 \to A1$ during the interaction interval—i.e. $M_{A0\to B0} > M_{A0\to A1}$. The energy required to transit to a different joint state comes from the heat reservoir, and excess energies that are gained due to a non-equilibrium system—e.g. transition to lower energy joint states, will be stored as work in the work reservoir. This non-equilibrium that depends on the symbols on the tape is the reason why the information ratchet is capable of functioning as a Maxwell's Demon.

These ratchet's energetics and transition probabilities of the system are fully captured in the Markov transition matrix M by Boltzmann statistics:

$$\begin{aligned} \frac{M_{x_{N+1}\otimes y'_N \to x_N \otimes y_N}}{M_{x_N \otimes y_N \to x_{N+1}\otimes y'_N}} &= e^{(E_{x_{N+1}\otimes y'_N} - E_{x_N\otimes y_N})/k_B T} \\ \implies \Delta E_{x_N\otimes y_N \to x_{N+1}\otimes y'_N} &= k_B T \ln \frac{M_{x_{N+1}\otimes y'_N \to x_N \otimes y_N}}{M_{x_N \otimes y_N \to x_{N+1}\otimes y'_N}} \end{aligned} \tag{2.4}$$

All Markov transition matrices M must satisfy equation 2.4 as the ratchet is thermalised with the heat reservoir, with their transition probabilities dependent on the energy difference of the transitions. The Markov transition matrix M is said to obey detailed balance, is reversible, and has a stationary distribution. Some papers such as [27], defined the individual energies of the ratchet states and bits—e.g. $E_0, E_1, E_A, E_B, E_{A0}, E_{B0}, \ldots$ etc. in their study. However, in this report, we will only define the Markov transition matrix M for the ratchet as it fully captures the energetics by equation 2.4. Each unique ratchets are then fully described by their unique transition matrix M.

As mentioned in the previous section, the dynamics of the ratchet's 'movement' along the tape is broken down into two parts: interaction transition and switching transition. The interaction transitions are governed by M and equation 2.4, and takes place in an interval of τ. This is when the ratchet interacts with the heat reservoir, extracting the heat Q necessary to make the transition to different energy states.

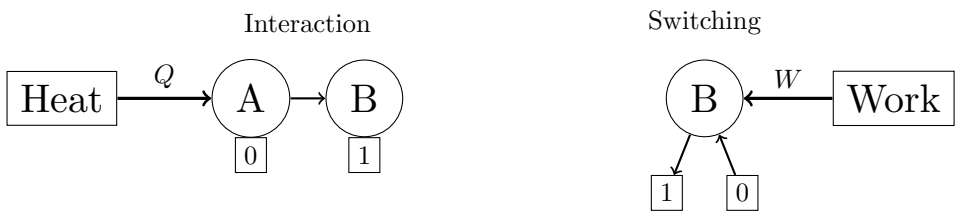
After the interaction transition, the ratchet now in state X_{N+1}, must detach from bit Y'_N, move to the next bit Y_{N+1}, and attach to it, forming the joint state $X_{N+1} \otimes Y_{N+1}$. This switching requires work to be done, essentially 'driving' the ratchet forward. During the switching transition, the ratchet only interacts with the work reservoir, extracting work W necessary for the switching of the bits. The interaction and switching transitions are illustrated in Figure 2.4. Therefore,

$$Q_N = E_{x_{N+1}\otimes y'_N} - E_{x_N\otimes y_N} \tag{2.5}$$

$$-W_N = E_{x_{N+1}\otimes y_{N+1}} - E_{x_{N+1}\otimes y'_N} \tag{2.6}$$

It is important to note that the system gains energy when it extract heat from the heat reservoir to make a transition, but the consumed work to make the switching are lost and dissipated back to the heat reservoir, hence the negative sign. Since this is a closed system, all the work that are needed to drive the ratchet comes directly from the heat reservoir with no input from external sources, making the ratchet autonomous. Thus, intuitively it is clear that the average or asymptotic rate of heat extraction is equal to the work production:

$$\langle W \rangle = \langle Q \rangle \tag{2.7}$$



$$X_N \otimes Y_N \to X_{N+1} \otimes Y'_N \qquad X_{N+1} \otimes Y'_N \to X_{N+1} \otimes Y_{N+1}$$
$$A0 \to B1 \qquad B1 \to B0$$

Figure 2.4: Interaction and switching transitions example. In the interaction transition, the ratchet extract heat to flip the bit from '0' to '1' and change its state from 'A' to 'B', $A0 \to B1$. This is followed by the switching transition, whereby the ratchet consume work to detach the bit '1' and attach the next bit '0', $B1 \to B0$.

The asymptotic rate of work production can be derived as:

$$\begin{aligned}
\langle W \rangle &= \lim_{N\to\infty} W_N \Pr(W_N) \\
&= \lim_{N\to\infty} \sum_{\substack{x_{N+1}, \\ y_{N+1}, y'_N}} (E_{x_{N+1}\otimes y'_N} - E_{x_{N+1}\otimes y_{N+1}}) \times \Pr(X_{N+1} = x_{N+1}, Y_{N+1} = y_{N+1}, Y'_N = y'_N) \\
&= \sum_{x',y'} E_{x'\otimes y'} \lim_{N\to\infty} \Pr(X_{N+1} = x', Y'_N = y') - \sum_{x,y} E_{x\otimes y} \lim_{N\to\infty} \Pr(X_{N+1} = x, Y_{N+1} = y)
\end{aligned} \tag{2.8}$$

Where in the second line, equation 2.6 was used. Here, x' is a dummy variable for its summation. The stationary distribution is defined as $\pi_{x\otimes y}$, and $\Pr(X_{N+1} = x', Y'_N = y')$ can be expressed in terms of $\Pr(X_N = x, Y_N = y)$:

$$\begin{aligned}
\lim_{N\to\infty} \Pr(X_N = x, Y_N = y) &= \lim_{N\to\infty} \Pr(X_{N+1} = x, Y_{N+1} = y) = \pi_{x\otimes y} \\
\Pr(X_{N+1} = x', Y'_N = y') &= \sum_{x,y} \Pr(X_N = x, Y_N = y) M_{x\otimes y \to x'\otimes y'}
\end{aligned}$$

Hence, the first and second terms in equation 2.8 are:

$$\begin{aligned}
\sum_{x',y'} E_{x'\otimes y'} \lim_{N\to\infty} \Pr(X_{N+1} = x', Y'_N = y') &= \sum_{x,y,x',y'} E_{x'\otimes y'} \pi_{x\otimes y} M_{x\otimes y \to x'\otimes y'} \\
\sum_{x,y} E_{x\otimes y} \lim_{N\to\infty} \Pr(X_{N+1} = x, Y_{N+1} = y) &= \sum_{x,y} E_{x\otimes y} \pi_{x\otimes y} \underbrace{\sum_{x',y'} M_{x\otimes y \to x'\otimes y'}}_{1} \\
&= \sum_{x,x',y,y'} E_{x\otimes y} \pi_{x\otimes y} M_{x\otimes y \to x'\otimes y'}
\end{aligned}$$

$\sum_{x',y'} M_{x\otimes y \to x'\otimes y'} = 1$ due to probability conservation. Therefore, the asymptotic work production rate is:

$$\begin{aligned}
\langle W \rangle &= \sum_{x,y,x',y'} (E_{x'\otimes y'} - E_{x\otimes y}) \pi_{x\otimes y} M_{x\otimes y \to x'\otimes y'} \\
&= k_B T \sum_{\substack{x,x'\in\mathcal{X} \\ y,y'\in\mathcal{Y}}} \pi_{x\otimes y} M_{x\otimes y \to x'\otimes y'} \ln \frac{M_{x'\otimes y' \to x\otimes y}}{M_{x\otimes y \to x'\otimes y'}}
\end{aligned} \tag{2.9}$$

Where equation 2.4 was used in the last line. It is clear that equation 2.9 is simply the weighted sums of the energy differences of all the joint state transitions at the stationary distribution.

2.4 Memory and Entropy Production

2.4.1 Tape Memory

We have yet to describe the workings of the tape of bits functioning as an information reservoir. Along with the ratchet's transition matrix M, the tape is another important variable in our study. It carries certain amount of information depending on its sequence of bits, which are typically measured by the Shannon entropy [35]. For a sequence of bits of length l, we have the Shannon block entropy:

$$H[Y_{0:l}] = -\sum_{w\in\{0,1\}^l} \Pr(Y_{0:l} = w) \log_2 \Pr(Y_{0:l} = w) \tag{2.10}$$

Where $\Pr(Y_{0:l} = w)$ is the probability of the sequence $Y_{0:l}$ having a word of w. In the perspective of communication theory, the ratchet acts as a communication channel, or a transducer, changing the information content of the tape after travelling along it. Thus, instead of imagining the ratchet moving along the tape, we can imagine sending an input tape $Y_{0:l}$ with a certain information content to the ratchet transducer, which then converts it into an output tape $Y'_{0:l}$ with a different information content. This is shown in Figure 2.5. This change in information content is what we are interested in.

Imagine an input tape sequence of all '0's, the entropy is thus minimum as knowledge of one bit gives us complete knowledge of the tape sequence with no uncertainty. Sending this input tape to a ratchet that functions as a Maxwell's Demon engine would then introduce uncertainty and thus entropy to the output tape, essentially 'dumping' the required entropic costs of the Maxwell's Demon to the tape.

Now, imagine another input tape generated with a fair coin flip—i.e. $\Pr(Y_N = 0) = \Pr(Y_N = 1) = 0.5$, the entropy is at the maximum with maximum uncertainty. Thus, we would expect the same ratchet to not be able to function as a Demon engine as it is not possible to further increase the entropy of the tape. However, by careful design of the ratchet's transition matrix M, we can instead turn the ratchet into an eraser, whereby work is consumed to reduce the entropy of the tape.

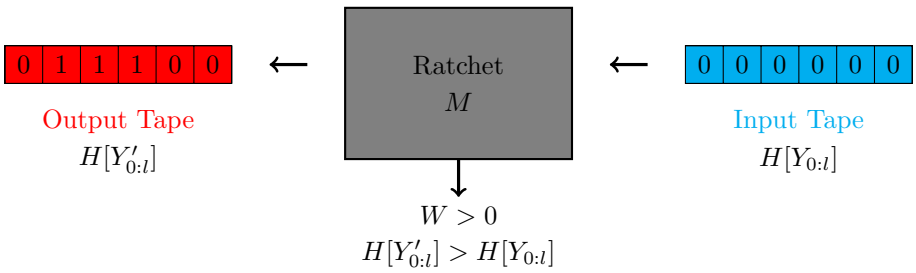


Figure 2.5: Ratchet as a transducer, changing the information contents of an input tape to an output tape, functioning as a Maxwell's Demon engine and producing work in the process.

These examples are extreme cases of maximum and minimum entropies. For general cases, we define a bias b in favour of bit '0', such that $\Pr(Y_N = 0) = b$. The entropy of a tape generated with a bias b would then be $H[b]$. What we are interested in is the relationship between the amount of work production and the amount of information that are being written or erased. This relationship is simply the modified Second Law that we see in equation 1.3 expressed in terms of work:

$$\begin{aligned}\langle W\rangle &\leq k_B T \ln 2\ \Delta H \\ &= k_B T \ln 2\ (H' - H[b])\end{aligned} \tag{2.11}$$

Where H' is the entropy of the output tape and H is the entropy of the input tape. However, this entropy measure is not complete. Consider instead an ordered input tape with sequence:

$$010101010101\ldots$$

On the one hand, this sequence gives a maximum fair coin flip entropy of $H[b = 0.5]$ as the number of bits '0' and '1' are the same. On the other hand, knowledge of a single bit will give us complete knowledge of the entire sequence as we know that the bits are alternating. In other words, there are temporal correlations or memory between the individual bits, such that we can predict future bits by looking at the past bits. In information theory, this means that the mutual information between past and future bits is non-zero [35]. This correlation is not captured by the Shannon entropy measure.

Thus, we require a new entropic measure known as the metric entropy rate or the Kolmogorov-Sinai entropy rate h_μ [36]–[38]. This entropy rate is able to take into account all temporal correlations and give the actual irreducible randomness per symbol. The previous Shannon entropy measure $H[Y_{0;l}]$ will be referred to as the Shannon block entropy, while h_μ will be referred to as the entropy rate.

$$\begin{aligned} h_\mu &= \lim_{l\to\infty} \frac{H[Y_{0:l}]}{l} \\ &= \lim_{l\to\infty} H[Y_l \mid Y_{0:l}] \end{aligned} \tag{2.12}$$

Where $H[X \mid Y]$ is the conditional entropy of X given Y [35]. Note that the subscript μ of the entropy rate has its roots in measure theory whereby a dynamical system can be described by a measure space with μ being its measure. In this case, the measure space is a probability space and has a total measure of unity. The subscript is only included for historical reasons and is insignificant here.

Thus, the new information processing relation is given in equation 2.13 with h'_μ denoting the entropy rate of the output tape. This is known as the Information Processing Second Law (IPSL). In later chapters, we will demonstrate that the IPSL is indeed a better bound compared to equation 2.11.

$$\begin{aligned} \langle W \rangle &\leq k_B T \ln 2 \ (h'_\mu - h_\mu) \\ &= k_B T \ln 2 \ \Delta h_\mu \end{aligned} \tag{2.13}$$

Temporal correlations are measured by the process' length-1 redundancy [35]:

$$\begin{aligned} H - h_\mu &= H - H[Y_N \mid Y_{0:N}] \\ &= \lim_{N\to\infty} I[Y_N; Y_{0:N}] \\ &\geq 0 \end{aligned} \tag{2.14}$$

Where $I[X;Y]$ is the mutual information between random variables X and Y. We have thus identified two main types of information tapes: memoryless (no correlations) and memoryful (with correlations). A memoryless tape simply has $h_\mu = H[b]$, and has entropy dependent on its statistical bias b of the bias coin flip. However, the entropy rate of a memoryful tape is much trickier, and requires a discussion on computational mechanics and Hidden Markov Models (HMM) [39], [40].

All types of input tape sequences can be generated with a finite-state HMM. HMMs are chosen as they are capable of generating all kinds of processes, including processes that are not produced by normal Markov chains. For a bias coin flip which is a memoryless input tape with statistical bias b, the Mealy's representation of the HMM [41] is shown in Figure 2.6.

The labels $0 : b$ and $1 : 1 - b$ means that there's a probability b of outputting '0' and a probability $1 - b$ of outputting '1' during the HMM state transition. The internal states of a HMM are known as the hidden states. A single hidden state will suffice for memoryless processes, but to generate the memoryful period-2 '01010101...' or '10101010...' processes, we requires more hidden states as shown in Figure 2.7.

Starting at hidden state D, there is a 0.5 probability of the first bit being '0' or '1', the rest of the bits will then alternates as the hidden states alternates between F and E. Thus, we can define a HMM with

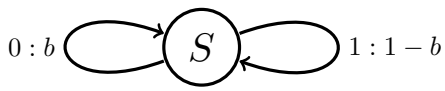


Figure 2.6: Hidden Markov Model generator with single hidden state S of a bias coin flip with probability b of outputting '0' and $1-b$ of outputting '1'.

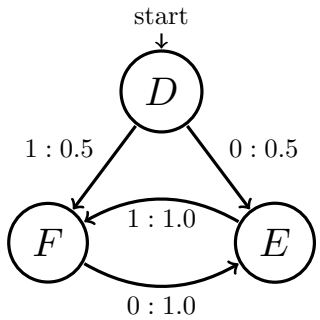


Figure 2.7: Hidden Markov Model generator of period-2 input tape with starting hidden state D.

a set of transition matrices with elements:

$$T^{(y_N)}_{s_N \to s_{N+1}} = \Pr(Y_N = y_N, S_{N+1} = s_{N+1} \mid S_N = s_N) \tag{2.15}$$

Where S_N denotes the hidden state at the N_{th} time step. $T^{(y_N)}_{s_N \to s_{N+1}}$ is the probability of transiting from hidden state $S_N = s_N$ to $S_{N+1} = s_{N+1}$ and outputting bit $Y_N = y_N$. For the period-2 process we have:

$$T^{(0)} = \begin{pmatrix} T^{(0)}_{D\to D} & T^{(0)}_{E\to D} & T^{(0)}_{F\to D} \\ T^{(0)}_{D\to E} & T^{(0)}_{E\to E} & T^{(0)}_{F\to E} \\ T^{(0)}_{D\to F} & T^{(0)}_{E\to F} & T^{(0)}_{F\to F} \end{pmatrix}, \quad T^{(1)} = \begin{pmatrix} T^{(1)}_{D\to D} & T^{(1)}_{E\to D} & T^{(1)}_{F\to D} \\ T^{(1)}_{D\to E} & T^{(1)}_{E\to E} & T^{(1)}_{F\to E} \\ T^{(1)}_{D\to F} & T^{(1)}_{E\to F} & T^{(1)}_{F\to F} \end{pmatrix}$$

$$T^{(0)}_{D\to E} = 0.5, \quad T^{(1)}_{D\to F} = 0.5, \quad T^{(0)}_{F\to E} = 1.0, \quad T^{(1)}_{E\to F} = 1.0$$

Sending a process or sequence generated by a HMM generator with transition matrices $T^{(0)}$ and $T^{(1)}$ to the ratchet, will give an output process or sequence generated by a HMM with different transition matrices $T'^{(0)}$ and $T'^{(1)}$ of elements:

$$\begin{aligned}
T'^{(y'_N)}_{s_N \otimes x_N \to s_{N+1} \otimes x_{N+1}} &= \Pr(Y'_N = y'_N, X_{N+1} = x_{N+1}, S_{N+1} = s_{N+1} \mid X_N = x_N, S_N = s_N) \\
&= \sum_{y_N} \Pr(Y'_N = y'_N, X_{N+1} = x_{N+1}, Y_N = y_N, S_{N+1} = s_{N+1} \mid X_N = x_N, S_N = s_N) \\
&= \sum_{y_N} \Pr(Y'_N = y'_N, X_{N+1} = x_N \mid Y_N = y_N, S_{N+1} = s_{N+1}, X_N = x_N, S_N = s_N) \\
&\quad \times \Pr(Y_N = y_N, S_{N+1} = s_{N+1} \mid X_N = x_N, S_N = s_N) \\
&= \sum_{y_N} \Pr(Y'_N = y'_N, X_{N+1} = x_N \mid Y_N = y_N, X_N = x_N) \\
&\quad \times \Pr(Y_N = y_N, S_{N+1} = s_{N+1} \mid S_N = s_N) \\
&= \sum_{y_N} M_{x_N \otimes y_N \to x_{N+1} \otimes y'_N} T^{(y_N)}_{s_N \to s_{N+1}} \\
&= \sum_{y_N} M^{(y'_N|y_N)}_{x_N \to x_{N+1}} T^{(y_N)}_{s_N \to s_{N+1}}
\end{aligned} \tag{2.16}$$

where we have defined

$$M^{(y'_N|y_N)}_{x_N \to x_{N+1}} = M_{x_N \otimes y_N \to x_{N+1} \otimes y'_N} \tag{2.17}$$

This new representation of M is used when we express our ratchet as a transducer to be consistent with the Mealy's representation of hidden states transitions, while the old representation is used in the context of Markov transitions of joint states. Since the output process is dependent on the input process and the ratchet's architecture, we would expect its generating HMM to have hidden states that are joint states of the input process' hidden states S_N and the ratchet states X_N, thus $T'^{(0)}$ and $T'^{(1)}$ has joint states of $S_N \otimes X_N$. Figure 2.8 shows the overall transducer picture of the information ratchet.

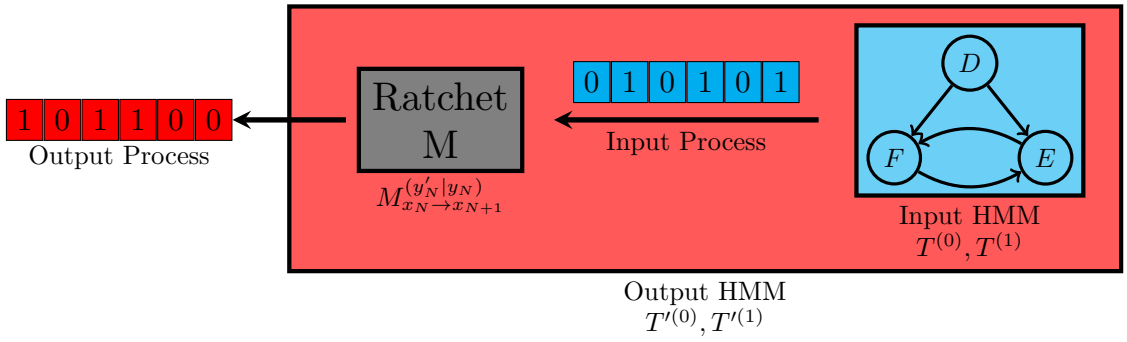


Figure 2.8: Overall transducer HMM picture of the information ratchet. The output HMM consists of both the input HMM with hidden states S_N and the ratchet with ratchet states X_N, thus the output HMM's hidden states are joint states of $S_N \otimes X_N$.

The output HMM's structure depends on the ratchet's structure. For example, given the ratchet Markov chain seen in Figure 2.2, and the period-2 input in Figure 2.7, we have the output HMM seen in Figure 2.9.

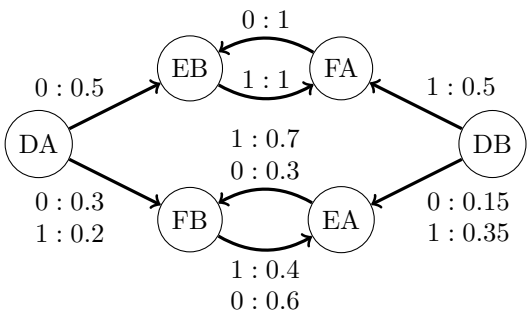


Figure 2.9: Output HMM generator of a system with a period-2 input tape, and a ratchet with the Markov chain in Figure 2.2. The hidden states of the output HMM are joint states of the input HMM's hidden states $\mathcal{S} \in \{D, E, F\}$ and ratchet states $\mathcal{X} \in \{A, B\}$.

The entropy rate of a sequence can be expressed in terms of the transition matrices $T^{(0)}$ and $T^{(1)}$ of the generating HMM [40]:

$$
\begin{aligned}
h_\mu &= \lim_{N\to\infty} H[Y_N \mid Y_{0:N}] \\
&= \lim_{N\to\infty} H[Y_N \mid S_N] \\
&= \sum_{s\in\mathcal{S}} \pi_s \lim_{N\to\infty} H[Y_N \mid S_N = s] \\
&= -\sum_{s\in\mathcal{S}} \pi_s \sum_{y\in\mathcal{Y}} T^{(y)}_{s\to s'} \log_2 T^{(y)}_{s\to s'}
\end{aligned} \tag{2.18}
$$

where π_s is the stationary distribution of the hidden states of the HMM. Similarly for the output process:

$$
h'_\mu = -\sum_{s'\in\mathcal{S}} \pi_{s'} \sum_{\substack{x,x'\in\mathcal{X},\\ y'\in\mathcal{Y}}} T'^{(y')}_{s\otimes x\to s'\otimes x'} \log_2 T'^{(y')}_{s\otimes x\to s'\otimes x'} \tag{2.19}
$$

2.4.2 Ratchet Memory

Similar to the tape of bits, the ratchet can also be categorised into memoryless and memoryful ratchet. In short, it is intuitive to see that ratchets that only has a single ratchet state are memoryless ratchets, whereas ratchets with more than one ratchet states are memoryful ratchets. To illustrate this, imagine a three states ratchet with states A, B, C. We can design the ratchet such that it only reaches state C after two consecutive '0' bits. The Markov chain for this ratchet is shown in Figure 2.10.

This means that whenever the ratchet is at state C, it must have processed two consecutive '0' bits and went through joint states $A0$ and $B0$. It is as though the ratchet 'remembers' or can determine the past states by looking at its current states, thus bearing memory. More formally, we say that there are mutual information between the current ratchet state and the past ratchet states. Even though this example was deliberately designed to demonstrate its memoryful properties, it is true for all ratchets with more than a single ratchet state, even if it is not obvious.

Single state ratchets on the other hand does not have this property. Without additional ratchet states, the ratchet cannot identify memories or temporal correlations in the input process—i.e. two consecutive '0' bits or period-2 inputs. The memoryless ratchet will only sees a period-2 input of 0101010 . . . as a fair coin flip, as temporal orders cannot be realised with only a single ratchet state.

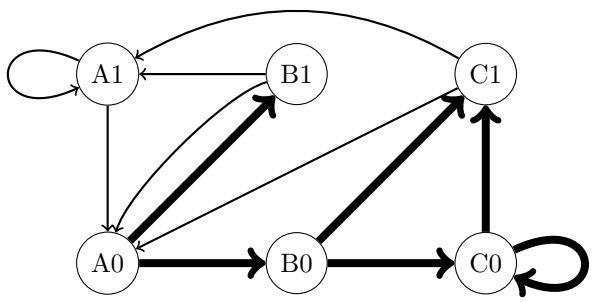


Figure 2.10: 3-states ratchet that can only reach ratchet state C after two consecutive '0' bits input. The bold arrows shows the transitions of '0' bits, whereas the non-bold arrows shows how the ratchet states return to state A if the input is '1' instead of '0', resetting the counting of '0' bits. Transition probabilities are omitted, but must summed up to 1 for outgoing edges.

We can draw parallels with the input process HMM by expressing the ratchet's Markov chain in Mealy's HMM transducer representation as shown in Figure 2.11. Similar to the input process HMM generator, the ratchet transducer requires more than one hidden state to be memoryful. The transducer representation is more general in computational mechanics, and its transition matrix M is expressed in the

form in equation 2.17. However, the Markov chain representation is useful for visualising the ratchet's functionality, as seen in Figure 2.10 whereby we can easily see that ratchet state C can only be reached after two consecutive '0' bits.

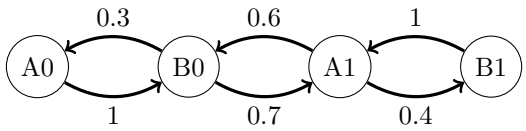
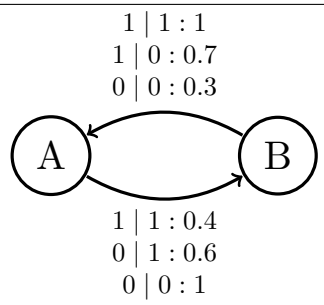
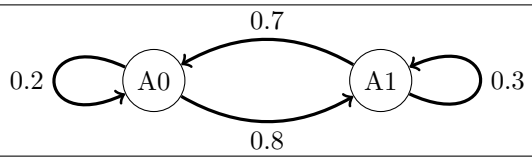
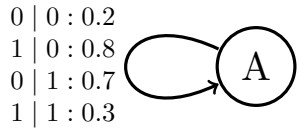
	Markov Chain	Transducer
Memoryful 2-States Ratchet		
Memoryless Ratchet		

Figure 2.11: Ratchet's Markov chain and transducer representation of arbitrary memoryful and memoryless ratchets.

Since both the input tape and the ratchet can be memoryful or memoryless, we can identify four different cases for the information ratchet system:

1. Memoryless Input, Memoryless Ratchet
2. Memoryless Input, Memoryful Ratchet
3. Memoryful Input, Memoryless Ratchet
4. Memoryful Input, Memoryful Ratchet

These cases are the starting point for our study, in the next chapter, we will explore each case in more details, except for the third case. As mentioned previously, a memoryless ratchet cannot tell the difference between a memoryless input and a memoryful input. Thus, the third case is essentially the same as the first case.

2.5 Numerical Program

As previously identified, the input tape and the ratchet are the only variables in our study, described by transition matrices $T^{(0)}, T^{(1)}$ and M respectively. This gave us a total of four main different cases, but with infinite number of different T and M. Thus, the aim of the program is to provide a simple framework to simulate the information ratchet system, enabling us to quickly prototype and experiment with input tapes of different information contents, as well as ratchets of different architectures.

We have written our program package in *Python*, with the only dependency being *numpy* [42]. Rather than simply computing the derived analytical equations, the program is an actual simulation of the system, such that each individual bit is simulated, went through, and changed according to transition matrix M, producing a new output tape. The program will then keep a list of all input and output joint states, for which the work done for each interval can be calculated from according to Boltzmann statistics in equation 2.4, thus making Boltzmann statistics the only physical law that the simulation obeys.

This allows us to emphasise that the physics of the system does not depend on the intricate derivations of computational mechanics or the subtle definitions of how the interaction and switching phases interacts with the heat and work reservoirs. They are simply useful tools for us to define the system analytically, as well as to make physical sense of the workings of the information ratchet.

The calculations of the output tape's entropy rate are also done by sampling the tape sequence directly, instead of relying on the derived transition matrices T'. While this is difficult for HMM which only reveals the output states—i.e. 001100010110..., it is possible in our case as we keep a list of the output joint states, enabling us to treat them as the hidden states of the output tape generator HMM. For example, if we have a output joint state transitions of:

$$B0\ B0\ A1\ B1\ B0\ A1\ B1\ A0\ B0\ B0\ A1\ B0\ A1\ B1\ B1\ A0$$

We can count the number of transitions:

$$\begin{aligned} A0 \to B0 &= 1, \quad A1 \to B1 = 3 \\ B0 \to B0 &= 2, \quad B1 \to A0 = 2 \\ B0 \to A1 &= 4, \quad B1 \to B0 = 1 \\ A1 \to B0 &= 1, \quad B1 \to B1 = 1 \end{aligned}$$

We can then build a HMM transition matrix by normalising the transition counts.

$$\therefore T'_{\text{counts}} = \begin{pmatrix} 0 & 0 & 0 & 2 \\ 1 & 2 & 1 & 1 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 3 & 1 \end{pmatrix} \quad \to \quad T' = \begin{pmatrix} 0 & 0 & 0 & 1/2 \\ 1 & 1/3 & 1/4 & 1/4 \\ 0 & 2/3 & 0 & 0 \\ 0 & 0 & 3/4 & 1/4 \end{pmatrix}$$

Since this T' was sampled from the output sequence itself, we can replace the actual generating transition matrices $T'_{s\otimes x \to s'\otimes x'}$ in equation 2.19 with it, and yet can expect to achieve the same h'_μ. However, the drawback to this is that the sampled transition matrix T' is only exact when the tape length approaches infinity.

On the user-level, these steps and calculations are defined as the methods of the tape and ratchet object classes. Users would only be required to import our package and call these methods to generate and simulate the system. The program will also identify wrong inputs by the users, such as matrices that does not have probabilities conservation, or non-square matrices.

To demonstrate the simplicity of our program, here is an example of an arbitrary 2-states ratchet with a period-2 input. Note that we will be taking computational units—i.e. $k_B = 1$ and $T = 1$. We have:

$$M = \begin{pmatrix} 0 & 0.3 & 0 & 0 \\ 1 & 0 & 0.6 & 0 \\ 0 & 0.7 & 0 & 1 \\ 0 & 0 & 0.4 & 0 \end{pmatrix}, \quad T^{(0)} = \begin{pmatrix} 0 & 0 & 0 \\ 0.5 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad T^{(1)} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0.5 & 1 & 0 \end{pmatrix}$$

First we import our program *inforatchet* as well as *numpy* as it will be useful in defining the transition matrices.

```
import numpy as np
import inforatchet.inputs
import inforatchet.ratchets
```

Next, we create the input tape with the 'BinaryTape' class with a length of 5000. and generate the period-2 sequence with transition matrices $T^{(0)}$ and $T^{(1)}$ of the generating HMM. Note that a longer

tape length will increase the accuracy of the results, but will also increase the program's runtime.

```
T0 = np.matrix([[0, 0, 0],
                [0.5, 0, 1],
                [0, 0, 0]])

T1 = np.matrix([[0, 0, 0],
                [0, 0, 0],
                [0.5, 1, 0]])

startState = [1, 0, 0]

inputTape = inforatchet.inputs.BinaryTape(5000)
inputTape.generateSequenceHMM(startState, T0, T1)
```

Here, we also specified 'startState' to specify a probability of 1 of the HMM starting at the first hidden state (recall in Figure 2.7 that the starting state is the first hidden state D).

Similarly for the ratchet, we simply need to create the 'FiniteStates' class with the transition matrix M, and the simulation can be started by specifying the input tape and the initial ratchet state.

```
M = np.matrix([[0, 0.3, 0, 0],
               [1, 0, 0.6, 0],
               [0, 0.7, 0, 1],
               [0, 0, 0.4, 0]])

ratchet = inforatchet.ratchets.FiniteStates(M)
outputTape, inStates = ratchet.startFixedTransitionRun(inputTape,
                                                       initRatchetState=0,
                                                       numTrans=1)
```

The run will return the 'outputTape' object which is the output tape, as well as 'inStates', recall our four different categorisations of joint state transitions in section 2.2, this is a list of all the input joint states $X_N \otimes Y_N$, whereas output joint states $X_{N+1} \otimes Y'_N$ are accessible from the 'outputTape' object. They will be needed for calculation of work later.

Note that we set 'initRatchetState=0' to set the initial ratchet state as 'A'. The program will automatically define 0=A, 1=B, 2=C, ... etc. depending on the size of the matrix M. We also specify the number of interactions in each interval with 'numTrans=1'—i.e. specifying the value of n in:

$$M^n \vec{\rho}(N) = \vec{\rho'}(N)$$

In our study, we will have $n = 1$, as is the case in equation 2.1. However, some studies such as [26], [43] assumes an interaction interval that consists of multiple transitions of the transition matrix—$n > 1$. [20] on the other hand, describes how for an extremely rapid self-equilibrating heat reservoir, the interaction transition can reach stationary distribution ($n \to \infty$) in a single interval, such that:

$$M\vec{\rho}_s = \vec{\rho}_s$$

where $\vec{\rho}_s$ is the stationary distribution of the ratchet transition matrix M. Our program is also capable of this by using the method 'startStationaryDistributionRun'.

Lastly, we can calculate the total work done W, the average work done $\langle W \rangle$, and the change in entropy rate Δh_μ. Note that the output joint states are called with 'outputTape.jointStates'.

```
# Calculate entropy rate
inputTape.calcEntropyRate()
outputTape.calcEntropyRate()
delta_h = outputTape.entropyRate - inputTape.entropyRate

# Calculate work
totalWork = ratchet.calcWork(inStates, outputTape.jointStates)
aveWork = totalWork / inputTape.length
```

To demonstrate the workability of our program, we need to compare our numerical results with the analytical solutions, as well as published results. We choose a system with a memoryless input generated with a statistical bias $b = 0.9$, with ratchet transition matrix of:

$$M = \begin{pmatrix} 0 & 1-p & 0 & 0 \\ 1 & 0 & q & 0 \\ 0 & p & 0 & 1 \\ 0 & 0 & 1-q & 0 \end{pmatrix}$$

p and q will be varied from 0 to 1, the work done and entropy rate for each value of p and q will be calculated and split into three categories:

1. Engine Regime: $k_B T \ln 2 \; \Delta h_\mu \geq \langle W \rangle > 0$
2. Eraser Regime: $0 > k_B T \ln 2 \; \Delta h_\mu \geq \langle W \rangle$
3. Dud Regime: $k_B T \ln 2 \; \Delta h_\mu \geq 0 \geq \langle W \rangle$

The first case are values of p and q whereby the ratchet is able to function as an engine, producing positive work, the second case are when it functions as an eraser, consuming work to erase information on the tape, giving a negative Δh_μ. The last case are when it neither functions as an engine nor an eraser. Note that in all cases, IPSL is obeyed as in equation 2.13.

We will plot p against q and colour code the three categories. Figure 2.12 shows the comparison between the numerical solutions from our program and the analytical solutions. This system was explored in [27] and a similar plot was obtained.

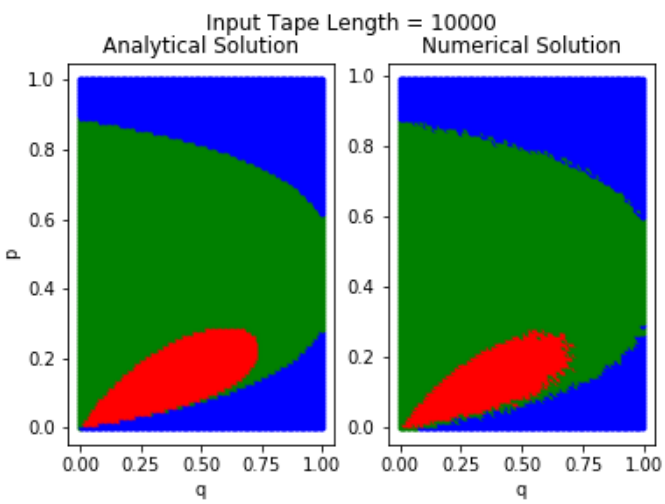


Figure 2.12: Comparison of analytical and numerical solution. Red regions indicates engine regime whereby work is positive, blue regions indicates eraser regime whereby work is negative, green regions indicates dud regime whereby the ratchet fails to function as both engine and eraser, such that work is negative but information difference is positive.

The numerical plot does not follow the analytical plot perfectly as our tape length does not approach infinity, and our arrays of p and q values are not fine-grained enough. However, this will suffice for this demonstration. Accuracy can be increased by increasing the tape length at the cost of computation time. Our study in later chapters are similar in spirit to this demonstration, such that a set of transition matrices M and T are specified, and work production and change in entropy rates are investigated. Note that for all future simulations in later chapters, we will use a tape length of 5000.

Having identified the different types of information ratchet systems, we have laid the groundwork for our study. There are two types of input tape processes and two types of ratchets. A memoryful input can only be leveraged by a memoryful ratchet, but how should we design the ratchet's structure to achieve maximum work production rate? Furthermore, for a memoryless input which can be processed by both a memoryful and a memoryless ratchet, which is the more efficient choice? In the next chapter, with the aid of the numerical program, we can attempt to answer these questions.

Chapter 3

Efficient Information Processing in the Information Ratchet System

3.1 Introduction

In the previous chapter, we have identified the four different cases for the information ratchet system:

1. Memoryless Input, Memoryless Ratchet
2. Memoryless Input, Memoryful Ratchet
3. Memoryful Input, Memoryless Ratchet
4. Memoryful Input, Memoryful Ratchet

We will group these cases into memoryful and memoryless inputs, and beginning our discussions with memoryful inputs first, before discussing the case of memoryless inputs. As mentioned, the third case of memoryful input with memoryless ratchet will be not discussed as it will yield the same result as the first case due to the inability of a memoryless ratchet to process a memoryful input process.

In the discussion of memoryful inputs, we will demonstrate why the Information Processing Second Law (IPSL) is the better bound, and is followed by a demonstration of an important principle for efficient memoryful information processing—synchronisation, which hints at a more fundamental principle of Requisite Complexity. On the other hand, the discussion of memoryless inputs will demonstrate that despite having the most efficient ratchet, there lies a bound beyond the Landauer limit that cannot be surpassed.

This chapter serves to show that efficient information ratchets require careful design for it to synchronise to the input process, as well as to identify the bound at which a maximum efficiency is reached.

3.2 Memoryful Inputs

3.2.1 Fully Correlated Inputs: Information Processing Second Law

In the previous chapter, we defined the two different relationships between the asymptotic work production rate and the change in entropy of the input tape, these are referred to as the modified Second Law bounds that takes into account informational entropies.

$$\begin{aligned}\langle W\rangle &\leq k_B T \ln 2 \ \Delta H \\ &= k_B T \ln 2 \ (H' - H)\end{aligned} \tag{3.1}$$

$$\begin{aligned}\langle W\rangle &\leq k_B T \ln 2 \ \Delta h_\mu \\ &= k_B T \ln 2 \ (h'_\mu - h_\mu)\end{aligned} \tag{3.2}$$

The second relation is the Information Processing Second Law (IPSL), we claimed that it is the better bound compared to the first relation, and it is only equal to the first relation if the input and output tapes are memoryless. Some works [44] has demonstrated how equation 3.1 can be violated. However, we will prove our claim by explicitly designing a system that violates the first bound but not the second bound. There are two ways to go about this.

First, we can design a ratchet that decreases temporal correlations of an arbitrary sequence, but increases its statistical bias—i.e. increasing entropy rate, but decreasing single-symbol entropy. These would result in $\Delta h_\mu > 0$, and $\Delta H < 0$. Thus, if the ratchet can extract positive work and function as an engine—i.e. $\langle W \rangle > 0 > \Delta H$, the first bound would be violated.

Second, we can simply choose an input sequence that is fully correlated. A fully correlated sequence has the process' length-1 redundancy at the maximum value of 1. Recall that the length-1 redundancy is the difference between the process' single-symbol entropy and its entropy rate, and is equal to the mutual information between the current bit Y_N and all the bits $Y_{0:N}$ before it.

$$\begin{aligned} H - h_\mu &= H - H[Y_N \mid Y_{0:N}] \\ &= \lim_{N\to\infty} I[Y_N; Y_{0:N}] \\ &\geq 0 \end{aligned} \tag{3.3}$$

Since H and h_μ are positive, a maximum length-1 redundancy means that the single-symbol entropy must be at the maximum $H = H_{\max} = 1$, while the entropy rate is at the minimum $h_\mu = h_{\mu_{\min}} = 0$. This is helpful to us as given a maximum $H_{\max}$, the output's single-symbol entropy can only be lesser or equal to it—i.e. $H' \leq H_{\max}$, thus giving $\Delta H \leq 0$. On the other hand, given a minimum entropy rate $h_{\mu_{\min}}$, the output's entropy rate can only be greater or equal to it—i.e. $h'_\mu \geq h_{\mu_{\min}}$, thus giving $\Delta h_\mu \geq 0$.

Therefore, if any arbitrary ratchet is capable of producing positive work, the first bound would be violated. One fully correlated input sequence is the familiar period-2 input, and a visualisation of the change in entropies are shown in Figure 3.1.

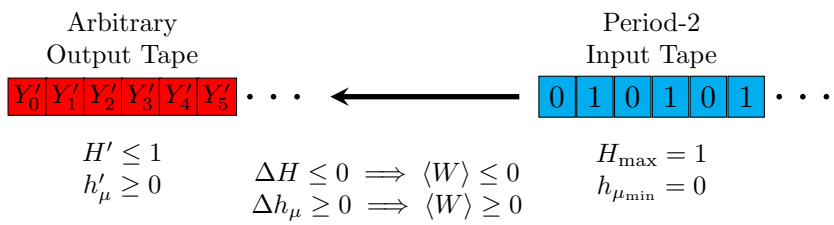


Figure 3.1: Change of information contents of a period-2 input tape. As single-symbol entropy is at the maximum $H_{\max} = 1$, the change in single-symbol entropy can only be negative, $\Delta H \leq 0$. Entropy rate on the other hand is at the minimum $h_{\mu_{\min}} = 0$, thus the change in entropy rate can only be positive, $\Delta h_\mu \geq 0$. Given these conditions, the first bound implies that work production must be negative, whereas the second bound implies that work production can be positive.

The ratchet must be memoryful, as mentioned in the previous chapter, a memoryful ratchet is required to process a memoryful input. Any arbitrary memoryful ratchet will suffice, provided that it is able to extract positive work. We will choose the simplest memoryful ratchet, which is a 2-states ratchet, and following the work of [27], we shall define the following ratchet transition matrix:

$$M = \begin{pmatrix} 0 & 1-p & 0 & 0 \\ 1 & 0 & q & 0 \\ 0 & p & 0 & 1 \\ 0 & 0 & 1-q & 0 \end{pmatrix} \tag{3.4}$$

Simulating this ratchet with the period-2 inputs resulted in two different plots shown in Figure 3.2 depending on the initial joint state of the system. If the initial joint states are $A0$ or $B1$, no positive work is done for all values of p and q. However, if initial joint states are $A1$ or $B0$, positive work (shown in red) is done for a wide array of p and q values. This shows that indeed, positive work can be extracted even with $\Delta H \leq 0$, violating the first Second Law bound which requires $\langle W \rangle \leq k_B T \ln 2 \, \Delta H$.

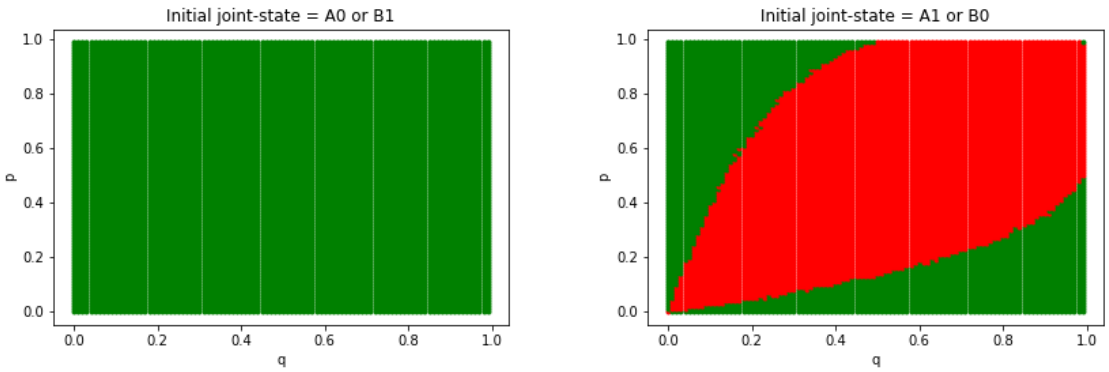


Figure 3.2: Ratchet's functionalities of different p and q values. Left – If initial joint state is A0 or B1. Right – If initial joint state is A1 or B0. Green areas: non-positive work. Red areas: positive work.

Since it is much easier to calculate the single-symbol entropy of a sequence, the first bound in equation 3.1 is more commonly used [43], [45], [46]. However, temporal correlations are the rule in nature, although a fully correlated environment like the period-2 input is rare, most system's information-bearing degree of freedoms have some temporal correlations to it, making the IPSL in equation 3.2 a better choice. In fact, it is difficult to produce a sequence that does not have correlations, this is most evident by the difficulty of producing true random numbers on a computer [47].

3.2.2 Synchronisation

Our demonstration of the IPSL being a better bound has revealed something interesting. As seen in Figure 3.2, the engine regime only appears for initial joint states $A1$ and $B0$, and no engine regime is present for initial joint states $A0$ and $B1$. Having two distinct functionalities might be useful in some cases, such as being able to switch a ratchet between engine or non-engine regimes by changing its ratchet states, but in other cases, this hinders the ratchet's functionalities.

To understand the cause of the two distinct ratchet functionalities, we look at the Markov chain of the ratchet transition matrix M, as shown in Figure 3.3. If the initial ratchet state is $X_0 = A$ and if the first bit of the tape is $Y_0 = 0$, thus $Y_0Y_1Y_2Y_3Y_4 \ldots = 01010\ldots$, we have the following transitions:

$$\begin{aligned}&(X_0 \otimes Y_0 \to X_1 \otimes Y_0') \to (X_1 \otimes Y_1 \to X_2 \otimes Y_1') \to (X_2 \otimes Y_2 \to X_3 \otimes Y_2') \to \ldots \\ &= (A0 \to B0) \to (B1 \to A1) \to (A0 \to B0) \to \ldots\end{aligned}$$

Whereby the intra-transitions within the brackets are interaction transitions and the inter-transitions between brackets are switching transitions. This gives an anticlockwise behaviour shown in red in Figure 3.3. The same occur for initial ratchet state $X_0 = B$, with initial bit $Y_0 = 1$, and recalling Boltzmann statistics, the work production for each interaction transition can be calculated as:

$$\Delta E_{A0 \to B0} = k_B T \ln \frac{1-p}{1} \leq 0, \quad 0 \leq p \leq 1 \tag{3.5}$$

$$\Delta E_{B1 \to A1} = k_B T \ln \frac{1-q}{1} \leq 0, \quad 0 \leq q \leq 1 \tag{3.6}$$

Both of which are negative, thus the absence of the engine regime as no positive work is done. We say that the ratchet is out of sync with the input process, and one way to identify this anticlockwise behaviour is when the input joint states before the interaction transitions are either $A0$ or $B1$.

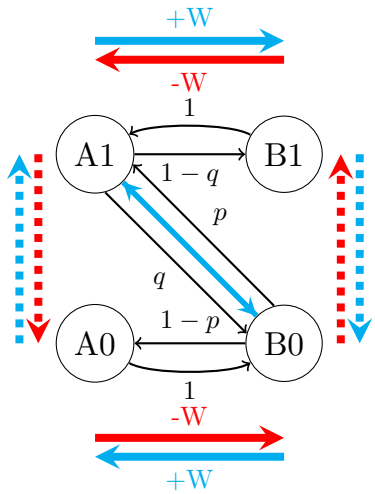


Figure 3.3: Markov chain of the 2-states ratchet with ratchet transition matrix M in equation 3.4. Cyan arrows shows the clockwise transitions whereby the ratchet is in sync with the input process. Red arrows shows the anticlockwise transitions whereby the ratchet is out of sync with the input process. Dashed arrows shows the switching transitions, and bold arrows shows the interaction transitions.

On the other hand, if the initial joint states are either $A1$ or $B0$, a clockwise behaviour (shown in cyan) is seen, with the following possible transitions and thus work productions:

$$\Delta E_{A1\to B1} = k_B T \ln \frac{1}{1-q} \geq 0 \tag{3.7}$$

$$\Delta E_{A1\to B0} = k_B T \ln \frac{p}{q} \tag{3.8}$$

$$\Delta E_{B0\to A0} = k_B T \ln \frac{1}{1-p} \geq 0 \tag{3.9}$$

$$\Delta E_{B0\to A1} = k_B T \ln \frac{q}{p} \tag{3.10}$$

Equations 3.7 and 3.9 are positives, whereas equations 3.8 and 3.10 are only positives for certain values of p and q. Thus, this clockwise behaviour produces positive works and we say that the ratchet is in sync with the input process. One way to identify this clockwise behaviour is when the input joint states before the interaction transitions are either $A1$ or $B0$.

If we want our ratchet to be functional as an engine, we must favour the clockwise regime where there are positive work production. For a non-anticipatory ratchet, such that the ratchet cannot predict the initial input, it cannot choose its initial ratchet state to be in sync with the input process. A random choice of the initial ratchet state would result in only 0.5 probability of being in the clockwise regime. However, one might notice that this notion of synchronisation of clockwise or anticlockwise behaviours seems to be a matter of ratchet design.

Thus, we will attempt to tweak our ratchet design such that the ratchet can transits from being out of sync to being in sync with the input process, the process of a ratchet synchronising itself to the input processes is known as self-synchronisation. A solution by [48] introduced a third ratchet state C, but before introducing a third state, we will see if we can do better by using only the current two ratchet states A and B. The new ratchet design is shown in Figure 3.4.

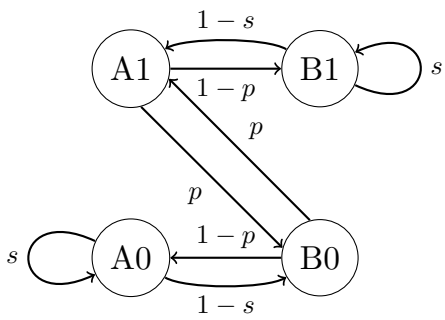


Figure 3.4: A ratchet that prevents or mitigates out of sync behaviours. This ratchet can self-synchronise itself to the clockwise regime whenever it is in the anticlockwise regime.

2-states Self-Synchronising Ratchet

We now have a ratchet that self-synchronises. If the initial joint state is either $A0$ or $B1$—i.e. anticlockwise regime, there is a probability s for the ratchet to remain at their current joint state, allowing the switching transition to bring it to be in the clockwise regime. Once in the clockwise regime, it will remain in the clockwise regime. Let us visualise this with an example. If the initial joint state is out of sync at $A0$, the following transitions can be seen:

$$(A0 \to B0) \to (B1 \to B1) \to (B0 \to A0) \to \dots$$

The first transition $A0 \to B0$ is an anticlockwise transition, which only happens with probability $1 - s$. The second transition $B1 \to B1$ is the synchronisation, which happens with probability s. This brings all subsequent transitions to be clockwise transitions, and in sync with the input process. One could imagine that this synchronisation by remaining in the current joint state is akin to a half-cycle phase shift by skipping a transition.

The ratchet transition matrix M_{sync} for this self-synchronising ratchet is:

$$M_{\text{sync}} = \begin{pmatrix} s & 1-p & 0 & 0 \\ 1-s & 0 & p & 0 \\ 0 & p & 0 & 1-s \\ 0 & 0 & 1-p & s \end{pmatrix} \tag{3.11}$$

The ratchet's thermodynamic functionalities for this 2-states self-synchronising ratchet are plotted in Figure 3.5.

For the case of initial joint states being out of sync (left of Figure 3.5), notice that there are no engine regimes for $s = 0$, this is because with $s = 0$, the ratchet cannot self-synchronise and thus remains in the anticlockwise regime, producing no positive work. Whereas, for the initial joint states that are already in sync (right of Figure 3.5), there are still engine regimes even for $s = 0$, as there is no need for self-synchronisation on a ratchet that is already synchronised. Another observation is that the engine regimes resides on the top left half of the plot, these are regions with $p > s$. Recall that the work production comes from the clockwise transitions of $B0 \to A0$ and $A1 \to B1$, and they have work production of:

$$\Delta E_{B0 \to A0} = \Delta E_{A1 \to B1} = k_B T \ln \frac{1-s}{1-p} \tag{3.12}$$

Thus, the work production is only positive for $p > s$, only around half of the spectrum of p and s values. Regardless, we have successfully designed a self-synchronising ratchet that is able to transits the ratchet from being out of sync to being in sync to extract positive work. We have done this without the need for a third ratchet state.

However, there is clearly a trade-off between synchronising rate and work production rate. This is

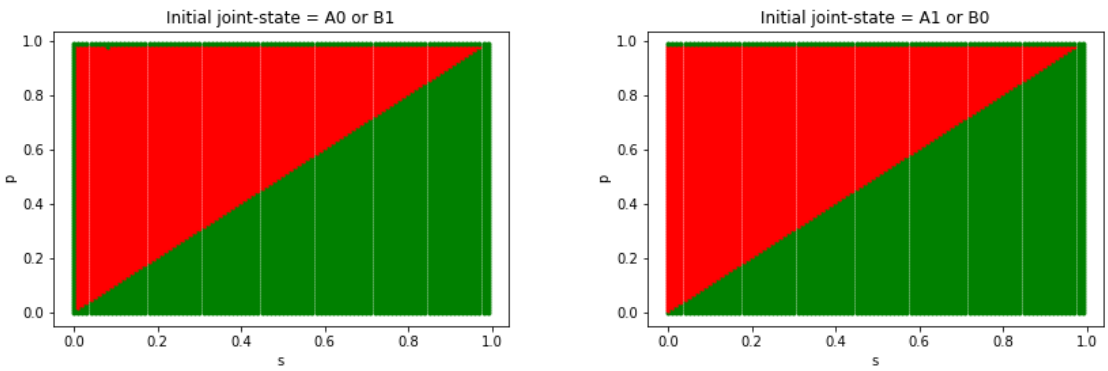


Figure 3.5: Ratchet's functionalities of the self-synchronising ratchet. Left – Out of sync initial joint states $A0$ or $B1$. Engine regimes (red) are present for $p > s$ with $s \neq 0$. For $s = 0$, there are no self-synchronisation and thus the ratchet remains out of sync and produce no positive work. Right – In sync initial joint states $A1$ or $B0$. Engine regimes are present even for $s = 0$, as it is already in sync with the input process and no synchronisation is needed.

evident in equation 3.12, whereby the work production for the transitions is dependent on the synchronising rate s. Specifically, if s is to increase, the work production decreases.

To make this trade-off more apparent, we fix the value of p to be dependent on s where $p > s$:

$$p = 1 - \frac{1-s}{e} \tag{3.13}$$

This choice is chosen such that the work production for the in sync transitions is $k_B T$:

$$\begin{aligned} \Delta E_{B0\to A0} = \Delta E_{A1\to B1} &= k_B T \ln \frac{1-s}{(1-s)/e} \\ &= k_B T \ln e \\ &= k_B T \end{aligned} \tag{3.14}$$

This gives us the asymptotic work production rate, which is simply the work production of each transitions multiplied by its probability of transition:

$$\langle W \rangle = k_B T \frac{1-s}{e} \tag{3.15}$$

[48] proposed a three-way trade-off between the heat absorbed during synchronisation Q_{sync}, the rate of synchronisation R_{sync}, and the work production rate $\langle W \rangle$, and is given by equation 3.16. A parallel was also made to the speed, cost, and fidelity of a computation [49].

$$Q_{\text{sync}} + k_B T \ln R_{\text{sync}} - k_B T \ln \left(1 - \frac{e\langle W \rangle}{k_B T}\right) = 0 \tag{3.16}$$

Our self-synchronising ratchet obeys this three-way trade-off. Along with the work production rate in equation 3.15, we have

$$R_{\text{sync}} = s \tag{3.17}$$

$$Q_{\text{sync}} = k_B T \ln \frac{s}{s} = 0 \tag{3.18}$$

The synchronising rate R_{sync} is the rate at which the synchronisation happens, recalling Figure 3.4,

this is the rate at which the transitions $A0 \to A0$ or $B1 \to B1$ happens. The heat absorbed during synchronisation Q_{sync} is the energy required for the synchronisation transitions $E_{A0\to A0}$ or $E_{B1\to B1}$.

Comparison with 3-states Self-Synchronising Ratchet

Before moving on, we should take a look at the 3-states self-synchronising ratchet proposed by [48]. A third ratchet state C, known as the synchronising state is added to the system, giving a Markov chain seen in Figure 3.6.

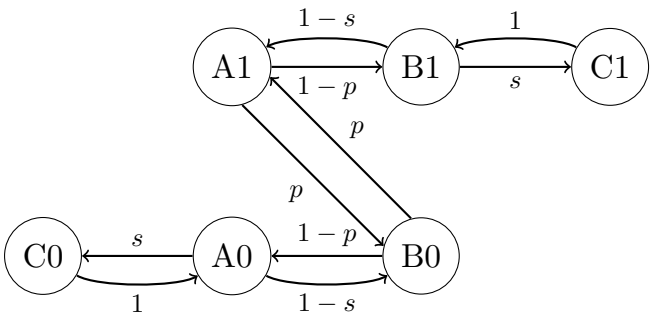


Figure 3.6: 3-states self-synchronising ratchet proposed by [48]. A third ratchet state is added that serves as the synchronisation state. The ratchet self-synchronise whenever it transits to ratchet state C, and will remain synchronised with the input process.

This 3-states self-synchronising ratchet work similar to our 2-states design. The difference is that instead of skipping a transition as we did, the third ratchet state C, acts as a 'bridge' or 'path' from the anticlockwise regime to the clockwise regime. We shall look at an example, if the ratchet is out of sync and has input joint state $A0$ or $B1$, there is a probability s (synchronisation rate R_{sync}) of transiting to ratchet state C. If the initial joint state is $A0$, this gives the transitions:

$$(A0 \to C0) \to (C1 \to B1) \to (B0 \to A0) \to (A1 \dots$$

The first transition $A0 \to C0$ is the synchronisation transition as the ratchet state transits to state C. The second transition $C1 \to B1$ is still part of the synchronisation as the ratchet state now exits from state C. From the third transition $B0 \to A0$, it now remains in the clockwise regime and will not enter state C again.

This seems to be slightly less efficient as two time steps are needed for synchronisation to take place, as compared to a single time step in our design. However, [48] points out that we can simply set the initial joint state to be at ratchet state C, such that synchronisation will always take place immediately. This is different from our design whereby synchronisation will only happen when the ratchet is out of sync. For this 3-states ratchet with initial ratchet state C, we have:

$$R_{\text{sync}} = M_{C0\to A0} = M_{C1\to B1} = 1 \tag{3.19}$$

$$Q_{\text{sync}} = \Delta E_{C0\to A0} = \Delta E_{C1\to B1} = k_B T \ln \frac{s}{1} \tag{3.20}$$

$$\langle W \rangle = k_B T \frac{1-s}{e} \tag{3.21}$$

Whereby for $\langle W \rangle$, we have again set the value of p to be dependent on s, as in equation 3.13. Again, the three-way trade-off of equation 3.16 is obeyed. If we set the initial ratchet state to be either A or B instead of C, we see that R_{sync} and Q_{sync} would change, but the three-way trade-off would still holds.

$$R_{\text{sync}} = M_{A0\to C0}M_{C1\to B1} = M_{B1\to C1}M_{C0\to A0} = s \tag{3.22}$$

$$\begin{aligned} Q_{\text{sync}} &= \Delta E_{A0\to C0} + \Delta E_{C1\to B1} \\ &= \Delta E_{B1\to C1} + \Delta E_{C0\to A0} \\ &= k_B T \ln \frac{1}{s} + k_B T \ln \frac{s}{1} \\ &= 0 \end{aligned} \tag{3.23}$$

Having the initial ratchet state to be A or B would result in a similar functionality as our 2-states self-synchronising ratchet, with the same R_{sync} and Q_{sync}. In this mode, if the synchronisation rate is low, it would take longer on average before the ratchet can synchronise with the input process, but there would be no energy cost for synchronisation. On the other hand, if the initial ratchet state is C, the ratchet will synchronise immediately, but at the expense of having energy costs to the synchronisation, this is because $Q_{\text{sync}} = k_B T \ln s \leq 0$.

One might then suggest that for both modes, we can set s to be large, such that for the mode with initial state of either A or B, the synchronisation happens faster, and for the mode with initial state of C, the energy costs are reduced. However, due to the three-way trade-off, this is compensated by a decrease in the work production rate as it is proportional to $1 - s$.

Figure 3.7 shows the plot of $\langle W \rangle$ against s for different values of p from 0.1 to 0.9 with a 0.1 increments. We see no major difference between our 2-states self-synchronising ratchet and the 3-states one proposed by [48], except at small values of s. The values of p are labelled beside each plot, and the domain of s for each p value is $0 < s \leq p$, this is because work production is only positive in this domain.

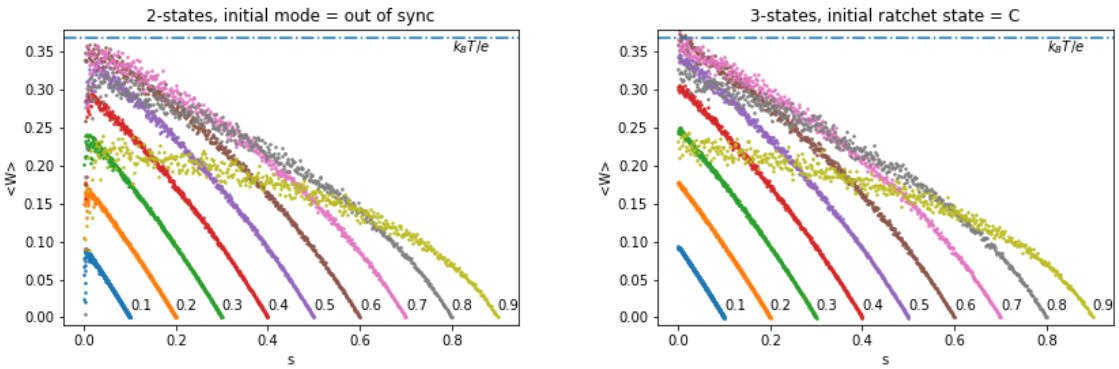


Figure 3.7: Work production rate $\langle W \rangle$ against s of the 2-states and 3-states self-synchronising ratchets for different values of p. The numbers at the bottom of each plot denotes the value of p. For each plot, only values of $s < p$ are plotted as only those values has positive work production. The dashed line denotes the value of $k_B T/e$, only a few stray data points exceed this value.

Interestingly, we see that for all p values, the work production rate is larger for smaller values of s. This make sense as the ratchets only need to go through synchronisation once. For the 3-states ratchet, the huge cost of Q_{sync} from a small s are only paid once, and thus is insignificant when compared to the gain in work production from the small s. However, for our 2-states ratchet, smaller values of s might cause the ratchet to only synchronise after a series of anticlockwise transitions has already taken place, resulting in lower work production rate for these small values of s.

Maximum work production rate is achieved for values of p between 0.6 and 0.7, but the work production rate does not exceed the value of $k_B T/e$, except for a few strays which we attribute to numerical errors from tape length limitations. We shall withhold the discussion of this limit to later sections.

Regardless, we see that the addition of the synchronising ratchet state C resulted in a better functionality than our 2-states ratchet. This is due to the ability to start the ratchet state at state C, ensuring synchronisation to take place immediately. On the other hand, for our 2-states ratchet, optimising the rate of synchronisation R_{sync} might be more advantageous than minimising the heat required for synchronisation Q_{sync} as the cost of Q_{sync} only happens once, whereas a slow R_{sync} increases the amount of time the ratchet is out of sync and producing negative work.

3.2.3 Noisy Inputs

However, in actual physical system, synchronisations can take place more than once. This is because in actual physical system, input processes are subjected to noises and fluctuations from the environment. Noises or fluctuations are disruptions to the patterns of a process, reducing the process' predictability and causing errors in the sequence. Thus a ratchet can becomes out of sync due to these errors. A self-synchronising ratchet would then need to recover from these occasional errors, but as the errors become more frequent, work production rate is sacrificed for synchronisation.

A period-2 input process alternates between bit 0 and 1. Errors can be added in the form of having a probability for a bit to repeat itself multiple times before alternating, causing a phase slip—e.g. 011010001010... . This can be implemented with the HMM generator shown in Figure 3.8, whereby there is a probability c for a bit to repeat itself, and a probability $1 - c$ to alternates. We would then expect that the range of values of p and s for engine regimes would decrease for increasing levels of errors. This is because the ratchet will return to the out of sync regime more frequently, and thus negative work production also happens more frequently.

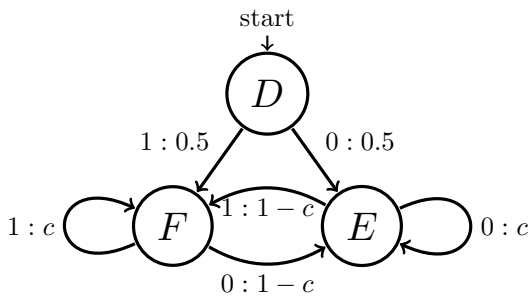


Figure 3.8: HMM generator of a noisy period-2 input process. Hidden states F and E now has a probability c of remaining in the same state and repeating the same output bit.

First, we shall look at the ratchet functionalities with increasing values of c. This is shown in Figure 3.9. We find no significant differences in terms of thermodynamic functionalities between our 2-states ratchet and the 3-states ratchet proposed by [48], both ratchets seems to have the almost the same arrays of p and s values to function as an engine. This might be because for subsequent desynchronises caused by these errors, the synchronisation rate and heat required are the same for both ratchets, and the slight advantage of the 3-states ratchet during the first synchronisation is insignificant.

As expected for both ratchets, the allowable values of p and s for engine regimes decreases as c increases. This trend continues to $c = 0.5$, whereby the input process is now equal to a fair coin flip, such that there is always a 0.5 probability of being bit 0 or 1, without any temporal correlations. This gives maximum single-symbol entropy and entropy rate, thus we would expect the ratchet to not be able to function as an engine. However, we does see a few engine regimes at $p \approx s$, which we attribute to numerical errors from a non-infinite tape, such that H and h_μ are only approximately maximal. Engine regimes starts showing up again for $c > 0.5$, and at $c = 1$, the input process is single bit, with a sequence of either 00000... or 11111..., giving minimum entropy rate, and thus engine regimes are expected again.

Different from an input with no errors, we can no longer assume that maximum work production rates are achieved with minimum synchronisation rates. This is because with errors, the ratchet will desynchronise frequently, and if synchronisation rate is slow, the ratchet cannot resynchronise with the process

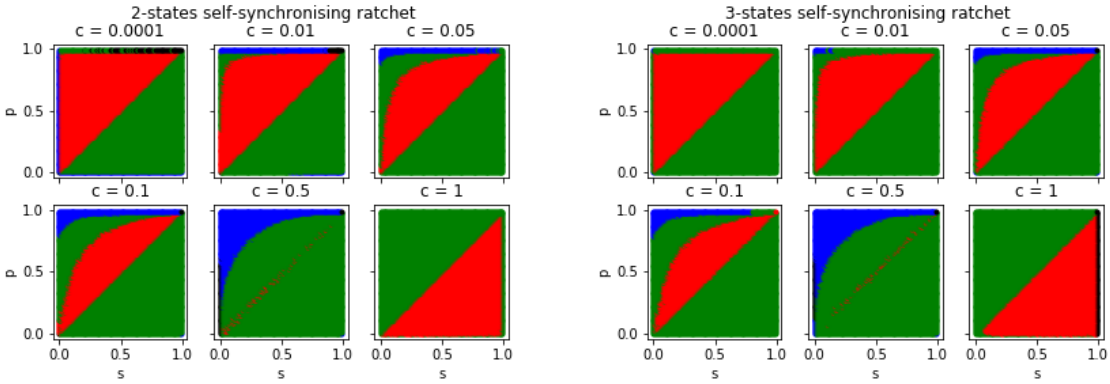


Figure 3.9: Ratchet functionalities of the 2-states (Left) and 3-states (Right) self-synchronising ratchets with different values of c.

fast enough before multiple negative work productions have already taken place. However, recall the fact that clockwise positive work production is proportional to $\ln \frac{1-s}{1-p}$, and thus a fast synchronisation rate would in turn lowers work production rate. Hence, this is essentially an optimisation problem, we want a synchronisation rate such that work production is maximum.

To visualise this, similar to the previous section, we will plot the work production rate $\langle W \rangle$ against the synchronising rate s, but this time we will set p to be a function of s as in equation 3.13, and plot different values of c instead. The plots are shown in Figure 3.10, and the values of c are labelled on the individual plots.

Quite different from the case of an input without errors shown in Figure 3.7, small values of s no longer give maximum work production rate. Rather, the work production rate reaches the minimum when $s \to 0$, as the ratchet is unable to synchronise with the input process, producing negative work. As expected, we see that as error rate c increases, synchronisation rate s must also increase for maximum work production rate. However, an overly large value of s than necessary would result in decreasing work production rate until it reaches $\langle W \rangle = 0$ at $s = 1$.

Quite obviously, work production rate decreases as error increases, this is because the ratchet desynchronises more often and thus produces more negative work. However, for $c = 0.25$, $c = 0.5$, and $c = 1.0$, work production rates are negative for all $s < 1$, and is only maximum at 0 when $s = 1$. Therefore, between $c = 0.1$ and $c = 0.25$, there must be a value c^* such that when $c > c^*$, no positive work is produced any more for all values of s.

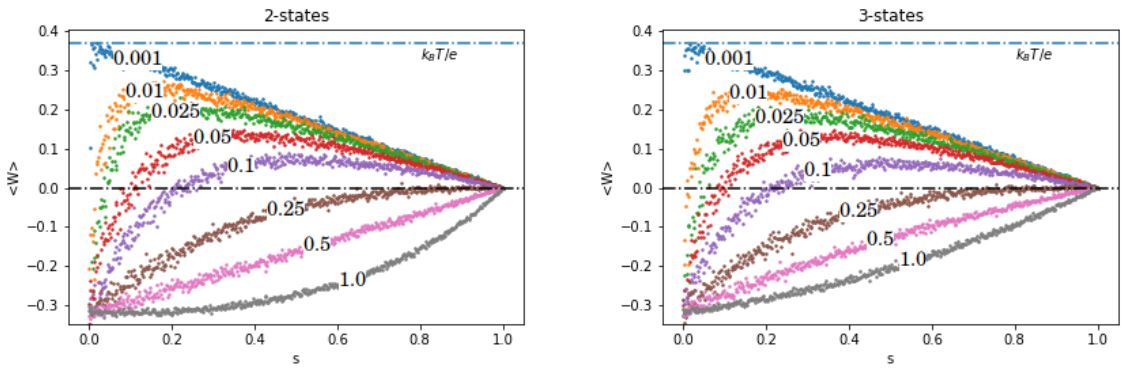


Figure 3.10: Work production rate $\langle W \rangle$ against synchronisation rate s of the 2-states and 3-states self-synchronising ratchets for different values of c. The numbers on each plot denotes the value of c.

We can then define c^* as the maximum error the input process can have before our ratchets are unable to extract any more positive work from it. A large value of c^* would means that the ratchet can still function as an engine in the presence of large input errors or environmental fluctuations, whereas a small value of c^* would means that the ratchet fails to function even with small input errors. Thus, c^* is a good measure of the self-synchronisation capabilities of a ratchet, and we would like to see how the 2-states and 3-states ratchet fare against each other.

In Figure 3.11, we plotted the maximum work production rate $\langle W \rangle_{\max}$ against c, and thus c^* is simply the value of c at which $\langle W \rangle_{\max}$ becomes 0. Interestingly, both the 2-states and 3-states ratchet has approximately the same value of c^*. This means that they are equally capable in functioning as an engine in the presence of input errors. This might be due to the fact that they both have the same synchronisation rate R_{sync} as well as heat required for synchronisation Q_{sync}. The value of c^* for the 3-states ratchet was derived to be $1/(1+e)$ in [48], and our numerical estimation for the 2-states ratchet gives $c^* \approx 0.266$ which is $\approx 1/(1+e)$.

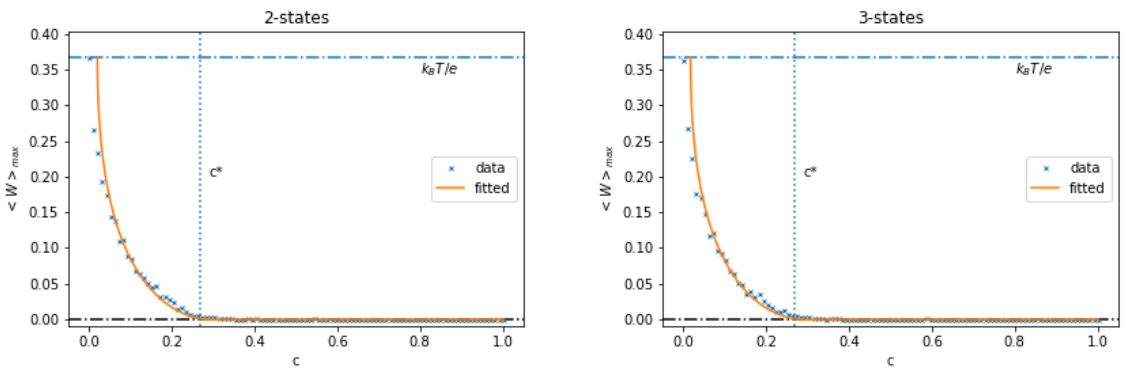


Figure 3.11: Maximum work production rate $\langle W \rangle_{\max}$ against c for the 2-states and 3-states self-synchronising ratchets. The vertical line c^* is the value of c such that $\langle W \rangle_{\max} = 0$ for all $c > c^*$. This is the maximum value of c before the ratchets are unable to produce positive work.

The decay of the maximum work production rate $\langle W \rangle_{\max}$ with increasing c seems to be an exponential decay. We can demonstrate this by plotting $\ln\langle W \rangle_{\max}$ against c as shown in equation 3.24, and attempting a linear fit. This is shown in Figure 3.12. We do note however that errors arises from taking the logarithm of $\langle W \rangle_{\max} \approx 0$, as the values approaches negative infinity. Therefore, the linear fitting is only done on data points with $c < c^*$.

$$\begin{aligned} \ln\langle W \rangle_{\max} &= A_1 c + A_2 \\ \implies \langle W \rangle_{\max} &= e^{A_2} e^{A_1 c} \end{aligned} \tag{3.24}$$

The values of A_1 and A_2 in equation 3.24 for both the 2-states and 3-states ratchets are found to be within the uncertainty of each other, allowing us to conclude that the 2-states and 3-states ratchets indeed functions the same. The values are shown in the table below, with $\bar{R}^2$ being the adjusted R-squared or coefficient of determination of the linear fit.

3.2.4 Requisite Complexity

Although these are examples that purposefully demonstrate the synchronisation properties of a ratchet, synchronisations are important for efficient ratchet designs in general. There are information-theoretic arguments for why a ratchet must be in sync with its input processes, which we will withhold until the next chapter.

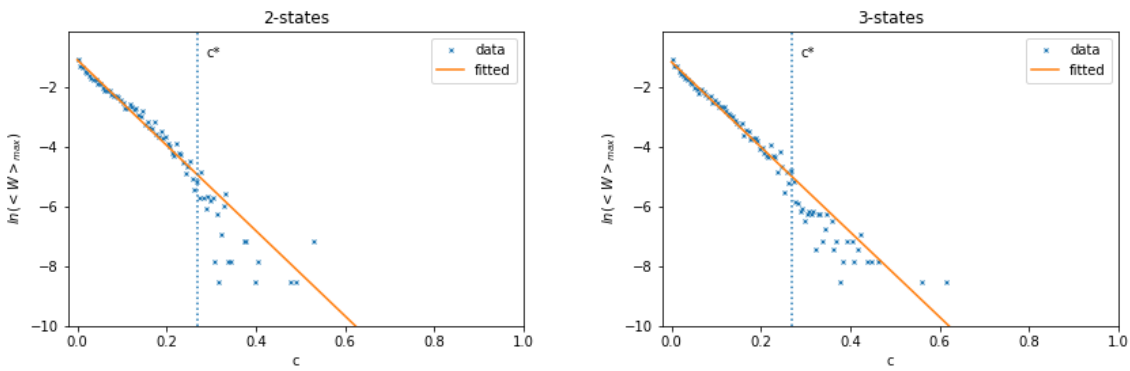


Figure 3.12: $\ln\langle W\rangle_{\max}$ against c plot. The relation is almost linear for $c < c^*$. This implies that $\langle W\rangle_{\max}$ decays exponentially with c. Note that majority of the data points beyond c^* are not shown as theoretically, their values approaches $\ln 0 = -\infty$. However, this is not completely true for our plot due to round off errors.

	2-states	3-states
A_1	-14.3 ± 0.4	-14.2 ± 0.3
A_2	-1.07 ± 0.06	-1.14 ± 0.05
$\bar{R}^2$	0.93857	0.95665

Table 3.1: Coefficients of equation 3.24 derived from linear fit.

The same authors who proposed the 3-states self-synchronising ratchet went on to propose a principle of Requisite Variety in their other works [50], drawing parallels with Ashby's Law of Requisite Variety in cybernetics [51], which states that the control system should have at least the same number of states as the system being controlled, for the whole system to be stable. The authors claimed that similarly, the ratchet must have at least the same number of ratchet states as the causal states of the input's ϵ-machine for efficient functionalities.

We have yet to define an ϵ-machine, but had in fact been using it [52]. A process can be generated by an infinite number of HMM, but its ϵ-machine is the minimal possible unifilar HMM to generate the process, and the hidden states of this minimal HMM are called the causal states. Figure 3.13 illustrates this for the period-2 input. Unifilarity refers to the uniqueness of the causal paths. For example, if a causal state can transits to two possible separate states, the output for the transitions must be different or unique. This is seen in the period-2 ϵ-machine in Figure 3.13, whereby the transitions away from causal state D outputs either 1 or 0 despite having the same probabilities, making each path unique and thus unifilar.

The principle of Requisite Variety is our motivation for attempting to design a 2-states self-synchronising ratchet and to compare it with the 3-states counterpart. As the period-2 process' generator has three causal states, the principle of Requisite Variety suggests that a 3-states ratchet is required to fully leverage the input process. However, we demonstrated that a 2-states ratchet can fulfils the same task with little differences.

To resolve this disagreement with Requisite Variety, [53] suggested that the principle is to be refined from one of variety to one of complexity, giving us the principle of Requisite Complexity instead. This new principle states that the ratchet must have at least the same uncertainty or complexity as the generating ϵ-machine to fully leverage the input process, or more generally—the complexity of the structure of a thermodynamically efficient system must match that of its environment. The reason our 2-states ratchet functions the same as the 3-states ratchet is simply because they have the same uncertainty or complexity:

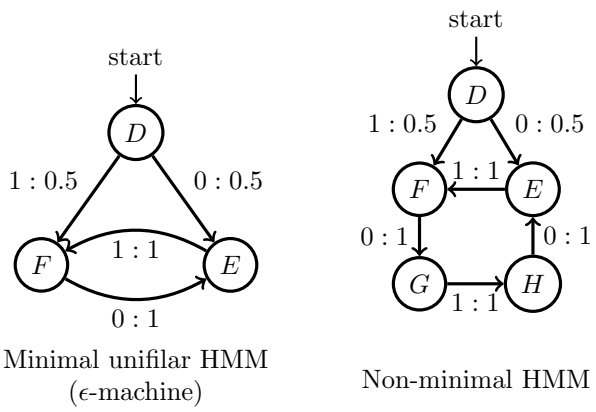


Figure 3.13: HMM generators of period-2 process. Left – The minimal possible unifilar HMM is known as the ϵ-machine, and its hidden states are referred to as the causal states. Right – The non-minimal HMM produces the same period-2 process, but it is not the most minimal possible HMM.

$$H[X_N] = -\sum_{xyx'y'} M_{x\otimes y \to x'\otimes y'} \log M_{x\otimes y \to x'\otimes y'} \tag{3.25}$$

The uncertainty of a process' ϵ-machine with causal states S_N is also known as the statistical complexity C_μ of the process:

$$H[S_N] = C_\mu \tag{3.26}$$

and thus the principle of Requisite Complexity states that for a ratchet to function as an efficient extractor of temporal correlations:

$$\begin{aligned} H[X_N] &\geq H[S_N] \\ &= C_\mu \end{aligned} \tag{3.27}$$

The principle of Requisite Complexity is the direct result of the need for the ratchet states to synchronise to the process' causal states. If equation 3.27 is not satisfied, the ratchet cannot synchronise to the input process. Thus, to design an efficient ratchet, the principle of Requisite Complexity must be taken into account. In the next chapter, we will look at the information-theoretic arguments for why this is the case.

3.3 Memoryless Inputs

Let us now step away from memoryful inputs and turn to memoryless inputs—inputs that has no temporal correlations, such that the length-1 redundancy is zero:

$$\begin{aligned} H - h_\mu &= 0 \\ h_\mu &= H \\ &= H[b] \end{aligned} \tag{3.28}$$

Where b is the statistical bias of input process. The statistical bias of a memoryless process is simply

the bias of a bias coin flip with single-symbol entropy:

$$H[b] = -b\log_2(b) - (1-b)\log_2(1-b) \tag{3.29}$$

Thus, while we are required to calculate or sample the entropy rate of a memoryful input process, all memoryless processes simply requires the statistical bias of the process, which is the ratio of the number of bit 0 over the whole length of the sequence. Equation 3.29 is plotted in Figure 3.14. Single-symbol entropy is maximum at $b = 0.5$—a fair coin flip, while it is minimum at $b = 0$ or $b = 1$, whereby the sequence consists of a sequence of the same symbols 00000... or 11111... .

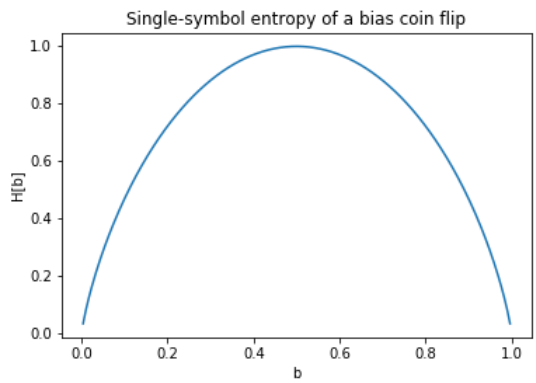


Figure 3.14: Single-symbol entropy $H[b]$ of a memoryless process against the statistical bias b.

The ϵ-machine of a bias coin flip consists of only a single causal state as shown in Figure 3.15.

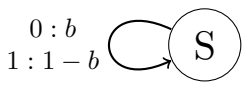


Figure 3.15: HMM generator for memoryless processes with single hidden state S. The bias b is in favour of bit 0, such that there is always a probability b of a bit being 0.

We would like to find the most efficient ratchet to extract work out of a memoryless process given a statistical bias b. If the principle of Requisite Complexity were to apply here, we only require a single state memoryless ratchet. However, Requisite Complexity does not set any constraints or limitations on the number of ratchet states one can use. Thus, in this section, we want to compare the efficiencies of a single state memoryless ratchet and a 2-states memoryful ratchet, given a memoryless input process.

3.3.1 Comparison between Memoryful and Memoryless Ratchets

First, for the memoryful ratchet, we will use the same 2-states ratchet seen in Figure 3.3. This is the 2-states ratchet without self-synchronisation, with the ratchet transition matrix given in equation 3.4. We do not have to worry about self-synchronisation here as the input process is now memoryless. This is because each individual bit is independent from each other, and depends solely on bias b. Thus, a ratchet cannot anticipate the next incoming bit to synchronise to it. Next, for the memoryless ratchet, there are only two joint states $A0$ and $A1$. Thus, there is only a single possible ratchet architecture for the memoryless ratchet shown in Figure 3.16.

$$M_{\text{2-states}} = \begin{pmatrix} 0 & 1-p & 0 & 0 \\ 1 & 0 & q & 0 \\ 0 & p & 0 & 1 \\ 0 & 0 & 1-q & 0 \end{pmatrix}, \quad M_{\text{1-state}} = \begin{pmatrix} 1-p & q \\ p & 1-q \end{pmatrix} \tag{3.30}$$

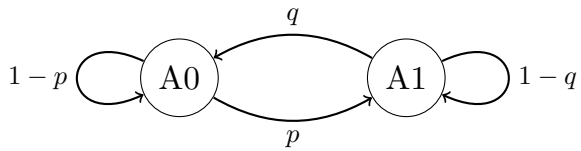


Figure 3.16: Markov chain of a single-state memoryless ratchet. There are only two joint states, resulting in a 2-by-2 matrix.

Similar to our analysis in previous sections, we will vary p and q, and plot the thermodynamic functionalities of the ratchet for different bias values from $b = 0.5$ to $b = 1.0$ with increments of 0.1. This is shown in Figure 3.17.

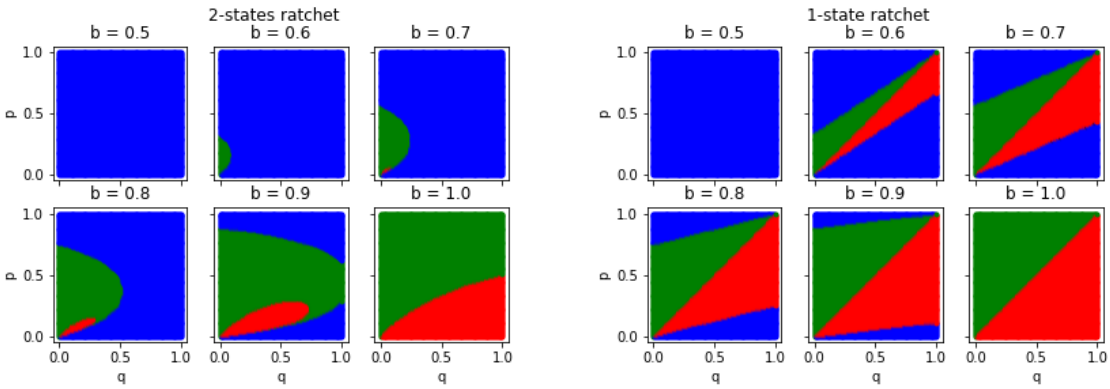


Figure 3.17: Thermodynamic functionalities of 2-states (Left) and 1-state (Right) ratchets with memoryless inputs of different biases. Red – engine regimes. Blue – eraser regimes. Green – dud regimes. Biases from 0.0 to 0.4 are omitted as they gives the same qualitative result as biases 1.0 to 0.6. This is due to the symmetry about $b = 0.5$, a bias of $b = 0.0$ is the same as $b = 1.0$ if we simply relabel bit 0 to bit 1.

The arrays of p and q that allows the ratchet to function as an engine increases with increasing bias, while eraser regimes increases with decreasing bias. This is what we would expect, recall the IPSL:

$$\begin{aligned} \langle W \rangle &\leq k_B T \ln 2 \ (h'_\mu - h_\mu) \\ &= k_B T \ln 2 \ (h'_\mu - H[b]) \end{aligned} \tag{3.31}$$

and our regimes classification:

1. Engine Regime: $k_B T \ln 2 \ (h'_\mu - H[b]) \geq \langle W \rangle > 0$
2. Eraser Regime: $0 > k_B T \ln 2 \ (h'_\mu - H[b]) \geq \langle W \rangle$
3. Dud Regime: $k_B T \ln 2 \ (h'_\mu - H[b]) \geq 0 \geq \langle W \rangle$

From Figure 3.14, at $b = 0.5$:

$$\begin{aligned} H[b = 0.5] &= H_{\max} = 1 \\ \therefore h'_\mu - H_{\max} &\leq 0, \quad h'_\mu \in [0, 1] \\ \implies \langle W \rangle &\leq 0 \end{aligned}$$

Thus, the ratchet can only function as an eraser at $b = 0.5$. On the other hand, as bias b increases, $H[b]$ decreases until the minimum value of 0 at $b = 1.0$:

$$\begin{aligned} H[b = 1.0] &= H_{\min} = 0 \\ \therefore h'_\mu - H_{\min} &\geq 0, \quad h'_\mu \in [0, 1] \\ \implies k_B T \ln 2 \ (h'_\mu - H[b]) &\geq 0 \end{aligned}$$

Thus, the ratchet can only function as an engine or as a dud at $b = 1.0$. Evident in Figure 3.17, this is true for both the memoryful and memoryless ratchets, as the IPSL applies for all cases. However, from the thermodynamic functionalities, the memoryless ratchet is clearly better at extracting work out of the memoryless input process due to the abundance of engine regimes as compared to the 2-states memoryful ratchet. This is due to the main difference between the 2-states memoryful and 1-state memoryless ratchets, which lies in the output process entropy rate h'_μ. We can give a qualitative argument on why the memoryless ratchet will be more efficient than the memoryful ratchet.

A memoryless ratchet cannot create temporal correlations in a process as it cannot store any information from the past in its single ratchet state. Since the input process has no temporal correlations, and the memoryless ratchet cannot create it, the output process must also be memoryless. Formally, we say that the ϵ-transducer of the single state ratchet acts as a memoryless channel for the input process [40], which according to the Data Processing Inequality [35], will reduce the temporal correlations in the input process. Thus, the length-1 redundancy of the output process must be lesser or equal to that of the input process:

$$H' - h'_\mu \leq H[b] - h_\mu \tag{3.32}$$

For memoryless inputs, the right hand side is 0, and since length-1 redundancies must be larger or equal to 0, the inequality reduces to an equality:

$$\begin{aligned} H' - h'_\mu &= 0 \\ h'_\mu &= H' \\ &= H[b'] \end{aligned} \tag{3.33}$$

Thus, the output process' entropy rate is simply the single-symbol entropy with b' being the statistical bias of the output process. Therefore, according to IPSL, the work production rate of the memoryless ratchet is bounded by:

$$\langle W \rangle_{\text{memoryless}} \leq k_B T \ln 2 \ (H[b'] - H[b]) \tag{3.34}$$

On the other hand, for a memoryful ratchet, it is possible to create temporal correlations in the output process as the ratchet can store past information in its ratchet states. This is because the ratchet state that the ratchet is currently in is dependent on past ratchet states, and the output process in turn depends on these ratchet states. Thus, the length-1 redundancy for the output process can be greater than 0:

$$\begin{aligned} H[b'] - h'_\mu &\geq 0 \\ H[b'] &\geq h'_\mu \\ \implies H[b'] - H[b] &\geq h'_\mu - H[b] \end{aligned} \tag{3.35}$$

Therefore, according to IPSL, the work production rate of a memoryful ratchet is lesser or equal to the

work production rate of a memoryless ratchet:

$$\begin{aligned}\langle W\rangle_{\text{memoryful}} &\leq k_B T \ln 2\ (h'_\mu - H[b]) && (3.36)\\ &\leq k_B T \ln 2 (H[b'] - H[b]) \\ &= \langle W\rangle_{\text{memoryless}} && (3.37)\end{aligned}$$

We can see this from the thermodynamic functionalities in Figure 3.17, the 1-state memoryless ratchet clearly has more engine regimes than the 2-states memoryful ratchet. We can also look at the maximum work production rate achieved by the ratchets at a fixed input bias value. This is shown in Table 3.2 for the memoryful ratchet, and Table 3.3 for the memoryless ratchet, where we can see that the maximum work production rates $\langle W\rangle_{\max}$ of the memoryless ratchet are much larger than that of the memoryful ratchet.

Bias	0.5	0.6	0.7	0.8	0.9	1.0
$\langle W\rangle_{\max}/k_B T$	0	0	0.000857	0.00563	0.0255	0.139
Efficiency at $\langle W\rangle_{\max}$	0	0	0.379	0.384	0.428	0.533

Table 3.2: Maximum work production rates and corresponding efficiencies for the 2-states memoryful ratchet with input processes of different bias values. Values are 0 for biases 0.5 and 0.6 as there are no engine regimes in those bias values.

Bias	0.5	0.6	0.7	0.8	0.9	1.0
$\langle W\rangle_{\max}/k_B T$	0	0.0183	0.0679	0.144	0.244	0.368
Efficiency at $\langle W\rangle_{\max}$	0	0.910	0.832	0.763	0.683	0.560

Table 3.3: Maximum work production rates and corresponding efficiencies for the 1-state memoryless ratchet with input processes of different bias values. Values are 0 for bias 0.5 as there are no engine regimes for bias 0.5.

We have defined efficiency as the achievability of the IPSL bound:

$$\text{Efficiency} = \frac{\langle W\rangle}{k_B T \ln 2\ \Delta h_\mu} \tag{3.38}$$

Now we can be quite convinced that a memoryless ratchet is the best choice at extracting positive work from a memoryless input. While we have provided a qualitative argument using the IPSL, and also verified it numerically, it is also possible to prove this using information-theoretical arguments. This is done so in [50] by using the equation for asymptotic work production rate that was derived in in the previous chapter:

$$\langle W\rangle = k_B T \sum_{\substack{x,x'\in\mathcal{X}\\ y,y'\in\mathcal{Y}}} \pi_{x\otimes y} M_{x\otimes y\to x'\otimes y'} \ln \frac{M_{x'\otimes y'\to x\otimes y}}{M_{x\otimes y\to x'\otimes y'}} \tag{3.39}$$

3.3.2 Achievability of Bounds

One intriguing observation that were hinted in our results but yet to be discussed is the fact that none of our ratchets can achieve a work production rate of more than $k_B T/e$. This is true regardless of whether we have a memoryful or memoryless input process, or whether our ratchet can self-synchronise efficiently. This is seen explicitly in Figure 3.7, 3.10, and 3.11, as well as in Table 3.3 whereby the greatest $\langle W\rangle_{\max}/k_B T = 0.368 \approx 1/e$.

Recall that the IPSL comes directly from the modified Second Law, which comes from Landauer's

Principle. The Landauer's Principle states that erasure of 1 bit of information requires at least $k_B T \ln 2$ of energy. This means that at $\Delta h_\mu = 1$, $\langle W \rangle \leq k_B T \ln 2$. However, the equality is not achieved, and this Landauer limit is not reached.

$$\begin{aligned} \langle W \rangle_{\max} &= \frac{k_B T}{e} \\ &< k_B T \ln 2 \end{aligned} \tag{3.40}$$

One could argue that for the case of memoryful input processes, that perhaps we have not designed a ratchet that fully synchronises efficiently with the input process to reach the bound. Thus, our motivation for comparing the memoryless and memoryful ratchets is to demonstrate that this maximum limit of $k_B T/e$ applies even to memoryless input processes, despite having the best ratchet choice of a memoryless ratchet.

We can solve the work equation and attempt to derive this limit. Referring to the memoryless ratchet transition Markov chain in Figure 3.16, work production occurs for joint state transitions of $A0 \to A1$, and $A1 \to A0$. The work production rate is simply the weighted sum of the work production of these transitions:

$$\begin{aligned} \langle W \rangle &= k_B T \left[\Pr(A0 \to A1) \ln \frac{M_{A1 \to A0}}{M_{A0 \to A1}} + \Pr(A1 \to A0) \ln \frac{M_{A0 \to A1}}{M_{A1 \to A0}} \right] \\ &= k_B T \left[pb \ln \frac{q}{p} + q(1-b) \ln \frac{p}{q} \right] \end{aligned} \tag{3.41}$$

Where

$$\begin{aligned} \Pr(A0 \to A1) &= \Pr(X_{N+1} = A, Y'_N = 1 \mid X_N = A, Y_N = 0) \\ &= \Pr(Y'_N = 1 \mid Y_N = 0) \end{aligned}$$

is the probability of finding the joint state at $A0$—$\Pr(X_N = A0) = \Pr(Y_N = 0) = b$, multiplied by probability of the transition $M_{A0 \to A1} = p$. Similarly for $\Pr(A1 \to A0)$, the probability of finding the joint state at $A1$ is $1 - b$ and the probability of transitioning to $A0$ is q.

Next, to find the $p_{\max}$ and $q_{\max}$ that gives the maximum work production rate, we take the partial derivatives and let them equal to 0:

$$\frac{\partial \langle W \rangle}{\partial p} = \frac{-b(p+q) + bp \ln \frac{q}{p} + q}{p} \tag{3.42}$$

$$0 = -b(p_{\max} + q) + bp_{\max} \ln \frac{q}{p_{\max}} + q$$

$$\implies p_{\max} = q \left[e^{\Omega\left(\frac{e(1-b)}{b}\right) - 1} \right] \tag{3.43}$$

$$\frac{\partial \langle W \rangle}{\partial q} = \frac{bp}{q} + (1-b) \ln \frac{p}{q} - 1 + b \tag{3.44}$$

$$0 = \frac{bp}{q_{\max}} + (1-b) \ln \frac{p}{q_{\max}} - 1 + b$$

$$\implies q_{\max} = p \left[e^{\Omega\left(\frac{eb}{1-b}\right) - 1} \right] \tag{3.45}$$

Whereby $\Omega()$ is the Lambert W function or the Omega function, such that given a function $f(z) = ze^z$,

the Omega function is defined as $\Omega(ze^z) = f^{-1}(ze^z) = z$. For $p_{\max}$, if we plot out the terms in the square brackets against b, we find that it is only valid for $0.5 \leq b \leq 1$, and it is maximum at 1 for $b = 0.5$. Whereas for $q_{\max}$, b must be $0 \leq b < 0.5$ for the terms to be valid, again at $b = 0.5$, it is maximum at 1. This is visualised in Figure 3.18.

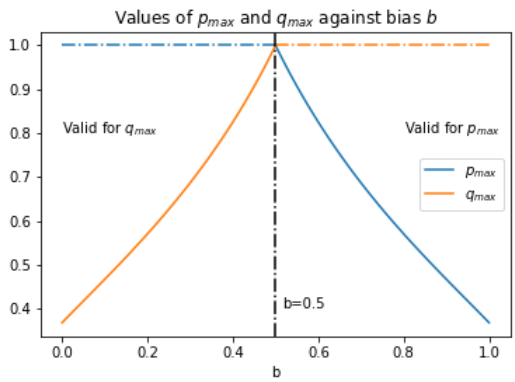


Figure 3.18: The plot is split at $b = 0.5$. For $b < 0.5$ on the left, $p_{\max} > 1$, but $q_{\max}$ is still valid. For $b \geq 0.5$ on the right, $q_{\max} > 1$, but $p_{\max}$ is still valid. Hence, we assume the equations for $p_{\max}$ and $q_{\max}$ is only valid in their domains of $0.5 \leq b \leq 1$ and $0 \leq b < 0.5$ respectively. The dashed lines are the values of q and p that we take in equation 3.43 and equation 3.45 respectively.

When $b < 0.5$, the value of $p_{\max}$ exceed the maximum possible value of 1. Similarly for $q_{\max}$, beyond $b > 0.5$, it exceeds the value of 1. Therefore, shown as dashed lines, we take their values to be 1 beyond their valid domains. Hence, the values of $p_{\max}$ and $q_{\max}$ that gives maximum work production rates are:

$$[p_{\max},\ q_{\max}] = \begin{cases} \left[1, \quad \left[e^{\Omega\left(\frac{eb}{1-b}\right)-1}\right]\right], & 0 \leq b < 0.5 \\ \left[\left[e^{\Omega\left(\frac{e(1-b)}{b}\right)-1}\right], \quad 1\right], & 0.5 \leq b \leq 1 \end{cases} \tag{3.46}$$

Therefore, the maximum work production rate is:

$$\langle W \rangle_{\max} = \langle W \rangle(b, p_{\max}, q_{\max}) \tag{3.47}$$

The corresponding change in entropy at $\langle W \rangle_{\max}$ is:

$$\Delta h_\mu(b, p_{\max}, q_{\max}) = \Delta H(b, p_{\max}, q_{\max}) = H[b'] - H[b] \tag{3.48}$$

Where b' is the statistical bias of the output process, which is the probability for the output joint states to output bit 0—i.e. probability for the interaction transitions $A0 \to A0$ and $A1 \to A0$:

$$b' = b(1 - p_{\max}) + (1 - b)q_{\max} \tag{3.49}$$

However, equation 3.48 is not the maximum possible change in entropy. The maximum possible change is:

$$\begin{aligned} \Delta h_{\mu_{\max}} = \Delta H_{\max} &= H_{\max} - H[b] \\ &= 1 - H[b] \end{aligned} \tag{3.50}$$

Now, we can plot and compare $\langle W \rangle_{\max}$, $\Delta H(b, p_{\max}, q_{\max})$, and $\Delta H_{\max}$. If the IPSL bound is achieved,

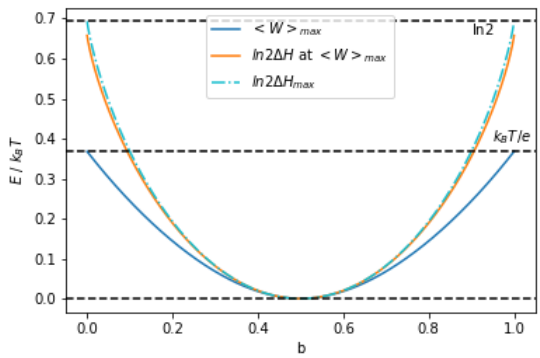


Figure 3.19: Plots of maximum work production rate $\langle W \rangle_{\max}$, its corresponding $\ln 2\ \Delta H$, and the maximum possible $\ln 2\ \Delta H_{\max}$ against input process bias b. Work production rate cannot exceed $k_B T/e$, and the Landauer limit of $\ln 2$ is not reached.

we would expect $\langle W \rangle_{\max}$ and $\ln 2\ \Delta H(b, p_{\max}, q_{\max})$ to align. If the ratchet fully leverage the information contents of the process, such that a maximum output entropy of $H' = H_{\max} = 1$ is achieved, we would expect $\ln 2\ \Delta H(b, p_{\max}, q_{\max})$ and $\ln 2\ \Delta H_{\max}$ to align. They are plotted in Figure 3.19.

We see that indeed work production rate (blue) cannot surpass $k_B T/e$. The change in entropy at maximum work production rate (orange) is also lower than the maximum possible change in entropy (cyan dashed). However, as bias value approaches $b = 0.5$, all three curves come closer to each other, which means that the ratchet is most efficient at smaller changes in entropy, and the IPSL bound is achieved.

Thus, the trade-off is that on the one hand, if we want an efficient ratchet that can fully leverage a memoryless process to extract work, such that the IPSL bound is achieved ($b \approx 0.5$), the work extracted will be small. On the other hand, if we want to extract maximum work ($b \approx 1$ or $b \approx 0$), not only is the IPSL bound not reached, the entropic change in the input process is also not the maximum possible.

The origin of this limit of $k_B T/e$ was suggested in [50], as the instantaneous and discontinuous changes in the system's Hamiltonian during the switching transitions, which resulted in a non-equilibrium distribution of the ratchet and bit states. Thus, the system have to relax towards Boltzmann distribution after every switching transitions, dissipating unrecoverable heat in the process.

Interestingly, shown in Figure 3.20, plotting the maximum work production rate $\langle W \rangle_{\max}$ against its corresponding change in entropy $k_B T \ln 2\ \Delta H(b, p_{\max}, q_{\max})$ shows that their relationship is a polynomial relation of degree 2. This equation allows us to compute the maximum work production rate $\langle W \rangle_{\max}$ directly from a given change in entropy ΔH.

$$\langle W \rangle_{\max} = A_1 (k_B T \ln 2\ \Delta H)^2 + A_2 (k_B T \ln 2\ \Delta H) + A_3 \tag{3.51}$$

The polynomial fit gives:

$$\begin{aligned} A_1 &= -0.4104 \pm 0.0009 \\ A_2 &= 0.8300 \pm 0.0005 \\ A_3 &= 0.00172 \pm 0.00005 \\ \bar{R}^2 &= 0.99987 \end{aligned}$$

Our results in this chapter shows that there is indeed a necessary structural condition for a ratchet to

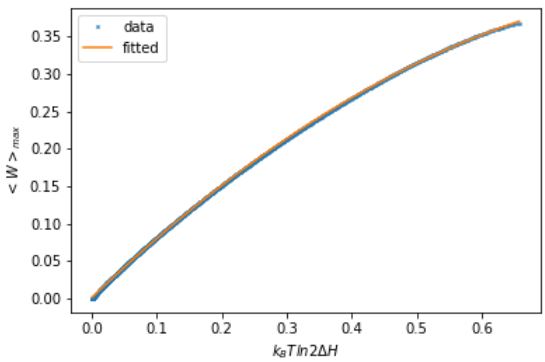


Figure 3.20: Plot of $\langle W \rangle_{\max}$ against $k_B T \ln 2\ \Delta H$, with polynomial fit.

efficiently extract work from a memoryful input process. This is the need for a ratchet to be able to synchronise to the input process—a ratchet must be able to transit to the right ratchet state in anticipation for the next incoming bit.

The ability for the ratchet to predict or anticipate the next incoming bit based on knowledge of the previous bit at each time step, means that there are correlations or mutual information between the ratchet states and the future inputs. Mutual information are shared information between systems, and are different from the internal memory that a memoryful ratchet holds. This is our first glimpse at the ratchet hoarding some amount of information that are unaccounted for. In this case, these information are useful for synchronisation, albeit only for the single time step that they exists in. The dissipations associated with the erasure of these information at the end of the time step are examples of the structural costs and dissipations of a ratchet's implementation, which we will explore in the next chapter.

We have also demonstrated that there is an unsurpassable bound $k_B T/e$ which is present in all our results. This bound is present even in a memoryless input and memoryless ratchet system, such that there are no unaccounted mutual information or correlations in the system. Therefore, we can associate this bound to be specific to the information ratchet system.

Chapter 4

The Costs of Information Processing

4.1 Introduction

In the previous chapter, we identified the key principle for information processing of a memoryful input process—synchronisation. A ratchet must be able to synchronise to the input process for efficient information processing, such that the least amount of heat is dissipated for the given computation or transduction of information. The principle of Requisite Complexity was also shown, as we demonstrated that a 2-states and 3-states self-synchronising ratchets are equally capable at self-synchronisation, both with and without environmental fluctuations or errors, this is due to them having the same statistical complexity, which is greater than that of the input process.

In this chapter, we will review information-theoretical arguments on the origins of the thermodynamic costs of information processing, these are the dissipations that are present in information processing. From these arguments, we will see that synchronisation and Requisite Complexity stems from the need to minimise these costs.

4.2 Transient Dissipation

One caveat about our previous discussions is the complete disregard for the information or entropy that the ratchet carries. One might notice this from the IPSL that Δh_μ is the only part of the equation that describes information, and it only consists of the change in entropy rate of the tape. However, a memoryful ratchet can of course hold memories about the past, and thus can also store and process information. This contribution from the ratchet was completely ignored by the IPSL. Here, we introduce a more general bound that includes the change in entropy of the ratchet:

$$\frac{\langle Q_{0\to N}\rangle}{k_B T \ln 2} \geq H[X_0, Y_0^G] - H[X_N, Y_N^G] \tag{4.1}$$

Equation 4.1 gives the heat dissipation of processing the input process from time step 0 to N, and it is bounded by the change in entropy of the entire joint system of ratchet and tape at time step 0 and N. Unlike the familiar IPSL, $\langle W\rangle$ is expressed as $-\langle Q\rangle$ here, as we are interested in the dissipation rather than the work production.

Note that the equation makes no distinction between input or output tape, Y_N^G denotes the entire global tape sequence at time step N, such that it includes the sequences that are already processed $Y'_{0:N}$, and the sequences that are not processed by the ratchet yet $Y_{N:\infty}$. Thus, while $H[Y_N]$ and $H[Y'_N]$ denotes the entropy of the input tape process and output tape process at time step N respectively, $H[Y_N^G]$ denotes the entropy of the entire tape without splitting it into input and output:

$$Y_N^G = Y'_{0:N} Y_{N:\infty} \tag{4.2}$$

Next, the familiar IPSL can be expressed into the same form as equation 4.1:

$$\lim_{N\to\infty} \frac{1}{N} \frac{\langle Q_{0\to N}\rangle}{k_B T \ln 2} \geq -(h'_\mu - h_\mu) \tag{4.3}$$

Recall that the IPSL bounds the asymptotic work production rate, whereas $\langle Q_{0\to N}\rangle$ describes the dissipation from time step 0 to N. Thus, a factor of $\lim_{N\to\infty} \frac{1}{N}$ is added to get the asymptotic rate.

As equation 4.1 accounts for the entire ratchet and tape system, and equation 4.3 only accounts for the tape, their difference is thus the dissipation that comes from the ratchet. We would expect that this dissipation is dependent on the ratchet's design, an efficient ratchet would have a smaller dissipation as compared to an inefficient ratchet. This dissipation is referred to as the transient dissipation [54]:

$$\begin{aligned}
\frac{\langle Q^{\text{tran}}_{0\to N}\rangle}{k_B T \ln 2} &\geq \lim_{N\to\infty} \left[\frac{\langle Q_{0\to N}\rangle}{k_B T \ln 2} - (-N(h'_\mu - h_\mu))\right] \\
&\geq H[X_0, Y_0^G] - H[X_N, Y_N^G] + N(h'_\mu - h_\mu) \\
&= H[X_0, Y_{0:\infty}] - H[X_N, Y'_{0:N}, Y_{N:\infty}] + N(h'_\mu - h_\mu) \\
&= \big(H[X_0] + H[Y_{0:\infty}] - I[X_0; Y_{0:\infty}]\big) \\
&- \big(H[X_N] + H[Y'_{0:N}, Y_{N:\infty}] - I[X_N; Y'_{0:N}, Y_{N:\infty}]\big) \\
&+ N(h'_\mu - h_\mu)
\end{aligned}$$

From the third line to the fourth line, the following identity was used:

$$H[A, B] = H[A] + H[B] - I[A; B]$$

Here, two assumptions will be made. First, we do not want arbitrary increase or decrease in dissipations that comes from different initial conditions—i.e. $H[X_0] - H[X_N] \neq 0$, thus the system is set to start at the stationary state, such that:

$$H[X_0] = H[X_N]$$

Second, the reasonable assumption is that the ratchet is non-anticipatory, such that the ratchet cannot predict or anticipate the incoming tape sequence by itself. This means that the initial ratchet state shares no information with the input tape sequence, or that they have no mutual information between them:

$$I[X_0; Y_{0:\infty}] = 0$$

Upon simplifications and expressing:

$$H[Y'_{0:N}, Y_{N:\infty}] = H[Y'_{0:N}] + H[Y_{N:\infty}] - I[Y'_{0:N}; Y_{N:\infty}],$$

$$H[Y_{0:\infty}] = H[Y_{0:N}] + H[Y_{N:\infty}]$$

most terms will cancels and gives:

$$\begin{aligned}
\frac{\langle Q^{\text{tran}}_{0\to N}\rangle}{k_B T \ln 2} &\geq H[Y_{0:N}] - I[Y_{0:N}; Y_{N:\infty}] - H[Y'_{0:N}] \\
&+ I[Y'_{0:N}; Y_{N:\infty}] + I[X_N; Y'_{0:N}, Y_{N:\infty}] + N(h'_\mu - h_\mu) \\
&= (H[Y_{0:N}] - Nh_\mu) - I[Y_{0:N}; Y_{N:\infty}] - (H[Y'_{0:N}] - Nh'_\mu) \\
&+ I[Y'_{0:N}; Y_{N:\infty}] + I[X_N; Y'_{0:N}, Y_{N:\infty}]
\end{aligned}$$

The limits of $N \to \infty$ are then taken on both sides, and defining the first and the third terms as the excess entropies of the input and output processes:

$$\begin{aligned} E &= \lim_{N\to\infty} \left(H[Y_{0:N}] - Nh_\mu\right) \\ &= \lim_{N\to\infty} I[Y_{0:N}; Y_{N:\infty}] \end{aligned} \tag{4.4}$$

$$\begin{aligned} E' &= \lim_{N\to\infty} \left(H[Y'_{0:N}] - Nh'_\mu\right) \\ &= \lim_{N\to\infty} I[Y'_{0:N}; Y'_{N:\infty}] \end{aligned} \tag{4.5}$$

and writing the dissipation as an asymptotic rate:

$$\langle Q^{\text{tran}} \rangle = \lim_{N\to\infty} \langle Q^{\text{tran}}_{0\to N} \rangle$$

we arrived at:

$$\frac{\langle Q^{\text{tran}} \rangle}{k_B T \ln 2} \geq -E' + I[Y'_{0:N}; Y_{N:\infty}] + I[X_0; Y'_{0:N}, Y_{N:\infty}]$$

$$\therefore \frac{\langle Q^{\text{tran}} \rangle_{\min}}{k_B T \ln 2} = -E' + I[Y'_{0:N}; Y_{N:\infty}] + I[X_0; Y'_{0:N}, Y_{N:\infty}] \tag{4.6}$$

For the third term, recall that this is a non-anticipatory ratchet:

$$\begin{aligned} I[X_0; Y_{N:\infty}] &= 0 \\ \implies I[X_0; Y'_{0:N}, Y_{N:\infty}] &= I[X_N; Y'_{0:N}] \end{aligned}$$

Note that we also relabelled X_0 as X_N. The second term in equation 4.6 vanishes as the output's past $Y'_{0:N}$ comes from the input's past $Y_{0:N}$, and has no correlations with the input's future $Y_{N:\infty}$, which are not processed by the ratchet yet.

$$\implies I[Y'_{0:N}; Y_{N:\infty}] = 0 \tag{4.7}$$

Therefore, we arrived at the main result for this section:

$$\frac{\langle Q^{\text{tran}} \rangle_{\min}}{k_B T \ln 2} = I[X_N; Y'_{0:N}] - E' \tag{4.8}$$

The minimum transient dissipation is the mutual information between the ratchet and the output's past, and is reduced by the output's excess entropy. This transient entropy production is the costs for a system to adapt to the environment [54]. The first term is also referred to as the implementation costs or dissipation:

$$\frac{\langle Q^{\text{impl}} \rangle_{\min}}{k_B T \ln 2} = I[X_N; Y'_{0:N}] \tag{4.9}$$

This is the costs of implementing a ratchet and will be different for different ratchet designs. Thus, an efficient ratchet must keep as little memory about the past outputs as possible to keep the mutual information low. We can interpret this as the ratchet keeping useless information about the past outputs that are unnecessary, creating and keeping useless entropy.

On the other hand, the output's excess entropy is a negative contribution to the dissipation, it is the mutual information between the output's past and the output's future. Thus, it is a measure of how predictable the future outputs are, given knowledge about the past outputs. To understand what the transient dissipation describes, we can visualise it by drawing an information diagram shown in Figure 4.1.

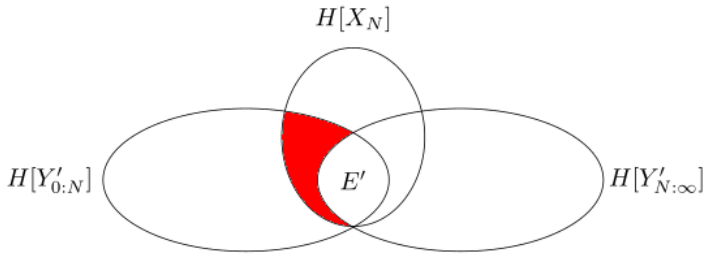


Figure 4.1: Information diagram of the ratchet and output process. E' is the mutual information between the output's past $Y'_{0:N}$ and output's future $Y'_{N:\infty}$. The red area shows the transient dissipation, it is the mutual information between the ratchet X_N and the output's past $Y'_{0:N}$ subtracted by E'.

The assumption for this information diagram is that the input process is memoryless. Recall from the previous chapter that this means all resulting correlations in the output process must be created by a memoryful ratchet, since the input process does not have any correlations to begin with. Therefore, the mutual information or excess entropy between the output's past and output's future, which describes their temporal correlations, must be fully captured by the ratchet. Hence the reason E' is contained within $H[X_N]$.

Therefore, the information diagram simply depicts an output process with temporal correlations that are 'bridged' by the ratchet, making the ratchet a generator of correlations, or simply a pattern generator. The transient dissipation in equation 4.8 is thus the heat dissipation required to generate this pattern from a patternless or memoryless input tape. It make sense that the excess entropy reduces the transient dissipation, as a large excess entropy means that we can predict more future outputs from the known past outputs, and in some sense, saving the ratchet of some work.

4.2.1 Pattern Generator

Note that the pattern generator is not a new functionality of the ratchet. A pattern generator simply processes a memoryless input process and outputs a memoryful output process. This is the case in the previous chapter whereby we compared the functionalities of a memoryless and a memoryful ratchet with a memoryless input process, and we demonstrated that memoryful ratchets are less efficient as it forms temporal correlations or patterns in the output process. It is less efficient as patterns generation consumes work.

With the help of the information diagram, and the transient dissipation in equation 4.8, let us find the most efficient pattern generator with the minimum transient dissipation. This is akin to minimising the red area in the information diagram. Pattern generators can be classified into two main types—predictive unifilar, and retrodictive co-unifilar [55].

Predictive Unifilar Ratchets

We shall look at the case of predictive unifilar transducers first. A unifilar transducer has its states R^+_N completely determined by the past outputs. Suppose we want to generate the period-2 patterns of 1010101..., the state R^+_{N+1} will be a function of the previous output—i.e. $R^+_{N+1}(Y'_N)$, such that if $Y'_N = 1$, the ratchet state R^+_{N+1} will be a state that outputs $Y'_{N+1} = 0$.

This means that in the information diagram, $H[R_N^+]$ is completely contained in the output past $H[Y'_{0:N}]$. This is depicted in Figure 4.2. Thus, the implementation cost is simply:

$$\frac{\langle Q^{\text{impl}}\rangle_{\min}}{k_B T \ln 2} = I[X_N = R_N^+; Y'_{0:N}] = H[R_N^+] \tag{4.10}$$

To minimise this, we require the most minimal ratchet that can still generate the pattern that we want, such that $H[R_N^+]$ is minimum. We denote this minimal ratchet as S_N^+ with:

$$\min(H[R_N^+]) = H[S_N^+] \tag{4.11}$$

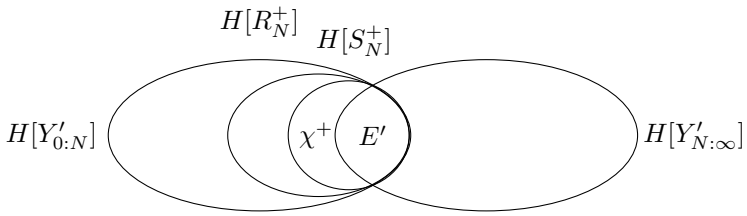


Figure 4.2: Information diagram with predictive unifilar ratchets R_N^+ and S_N^+. The ratchet's entropy atom is contained in the output's past $H[Y'_{0:N}]$ as it is a function of it. S_N^+ has the most minimal transducer, having a smaller area than the non-minimal R_N^+.

Since the inputs are memoryless, the most minimal ratchet that generates a pattern is simply the ϵ-machine of the pattern. Recall that the entropy of an ϵ-machine is also known as the statistical complexity C_μ. Thus, the implementation cost is simply the statistical complexity of the generated output pattern C'_μ:

$$\begin{aligned}\frac{\langle Q^{\text{impl}}\rangle_{\min}}{k_B T \ln 2} = I[S_N^+; Y'_{0:N}] &= H[S_N^+] \\ &= C'_\mu\end{aligned} \tag{4.12}$$

Therefore, to get the minimum transient dissipation, the output excess entropy E' is simply subtracted. This is also know as the crypticity χ^+ of the pattern [35]:

$$\begin{aligned}\frac{\langle Q^{\text{tran}}\rangle_{\min}}{k_B T \ln 2} &= C'_\mu - E' \\ &= \chi^+\end{aligned} \tag{4.13}$$

Crypticity as its name implies is a measure of how cryptic a process is. For example, recall the ϵ-machine of the period-2 process. The probability of causal state E and F are $\Pr(S_N = E) = 0.5$ and $\Pr(S_N = F) = 0.5$, this gives a statistical entropy of $C_\mu = H[0.5] = 1$. Additionally, knowledge of just one past bit—i.e. $Y'_0 = 1$, gives us complete knowledge of future bits $Y'_{1:\infty} = 01010...$, giving us an excess entropy of $E' = I[Y'_{0:N}; Y'_{N:\infty}] = 1$. Therefore, the crypticity χ of the period-2 process is 0 as there is nothing cryptic about a process that we can fully predict. Thus, according to equation 4.13, no heat dissipation is required to generate the period-2 pattern.

On the other hand shown in Figure 4.3, a golden mean process that generates sequences of bit 0 separated by multiple bit 1 has $\Pr(S_N = A) = 2/3$ and $\Pr(S_N = B) = 1/3$, giving $C_\mu = H[2/3] \approx 0.918$. However, the excess entropy is $E' \approx 0.252$, giving us a non-zero crypticity. This is because we cannot fully predict the future output from the past due to a 0.5 probability of the bit 1 repeating. Thus, according to equation 4.13, a heat dissipation of $k_B T \ln 2\ \chi$ is needed to generate this pattern.

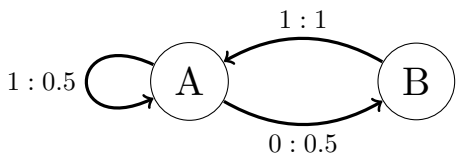


Figure 4.3: Minimal unifilar generator (ϵ-machine) of the golden mean process. The golden mean process has bit 0 separated by one or more bit 1—e.g. 0110101110... .

Retrodictive Co-unifilar Ratchets

The second class of pattern generators is the retrodictive co-unifilar transducers. Retrodictive implies that the ratchet state R_N^- is a function of the future output. Although this might seems unintuitive, it simply means that we can only tell what state a transducer is in, after looking at future outputs. This is the consequence of co-unifilarity, which is also referred to as non-unifilarity. Figure 4.4 depicts the retrodictive minimal co-unifilar generator of the same golden mean process.

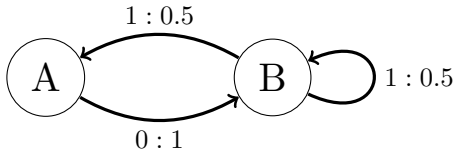


Figure 4.4: Minimal co-unifilar generator of the golden mean process.

Imagine observing a sequence of 01011 from the retrodictive minimal co-unifilar golden mean generator. The last output of $Y_4' = 1$ does not give us any information on the state S_4^-, as both transitions $S_3^- \to S_4^- = B \to B$ and $S_3^- \to S_4^- = B \to A$ gives us an output of bit 1. It is only through knowledge of the next bit—i.e. $Y_5' = 0$, then can we know the previous state to be $S_4^- = A$, since output of bit 0 is only possible for the transition $A \to B$. If instead $Y_5' = 1$, then we know that $S_4^- = B$, since only transitions from state B outputs bit 1.

On the other hand, for the previous case of an ϵ-machine in Figure 4.3, an output of bit 1 means that the generator is in causal state $S_N^+ = A$ as only transitions of either $B \to A$ or $A \to A$ will give an output of bit 1. Similarly for the case of output bit 0 giving us knowledge that the causal state is $S_N^+ = B$. Note that a minimal co-unifilar generator cannot be labelled as an ϵ-machine as the requirement for ϵ-machines is unifilarity.

Drawing out the information diagram for the retrodictive co-unifilar ratchet (Figure 4.5) require little discussion. It is clear from the diagram that the red areas in Figure 4.1 are completely eliminated. This is because the ratchet's entropy being functions of the future output, are fully contained in $H[Y'_{N:\infty}]$.

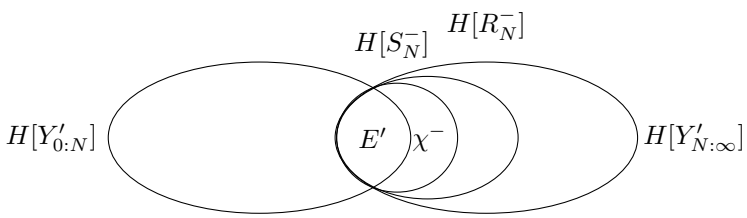


Figure 4.5: Information diagram with retrodictive co-unifilar ratchets R_N^- and S_N^-. The ratchet's entropy atom is contained in the output's future $H[Y'_{N:\infty}]$ as it is a function of it.

Therefore, we have:

$$\begin{aligned}
\frac{\langle Q^{\text{impl}}\rangle_{\min}}{k_B T \ln 2} &= I[X_N = R_N^-; Y'_{0:N}] \\
&= E' \\
\implies \frac{\langle Q^{\text{tran}}\rangle_{\min}}{k_B T \ln 2} &= E' - E' \\
&= 0
\end{aligned} \tag{4.14}$$

Amazingly, retrodictive pattern generators requires no transient dissipation. This is true regardless of the statistical complexity of the pattern to be generated, and is true regardless of the size of $H[R_N^-]$, such that we do not need the generator to be minimal. Therefore, while some works has claimed that simple is better for pattern generators [56], we have demonstrated that it is only true for predictive unifilar generators. Instead, the most efficient generator are retrodictive generators, and simplicity or minimality does not affect this efficiency.

4.3 Modularity Dissipation

Unfortunately, our discussion on transient dissipation only clarified the ratchet's relation to the output process, describing the mechanism for efficient pattern generation at the output process. This is evident by the lack of input processes $Y_{0:N}$ or $Y_{N:\infty}$ in the derived equations. The principle of Requisite Complexity, and the need for a ratchet to synchronise to the input, are both statements on how the ratchet interacts with the input process, which were not explained by transient dissipation. Thus, we will now introduce another form of dissipation that relates the ratchet and the input process.

This dissipation is the dissipation due to modular information processing. It has been argued in [53] that information processing on individual parts of the system instead of the whole, contributes to entropy production known as the modularity dissipation. Modularity is actually the norm in physical systems, such as neurons in the brain, and digital logic gates. In these systems, individual areas does their own information processing, working together to complete complex tasks and computations. This is also true for our information ratchet system, as during each interaction interval, we are only changing a single bit of the whole information reservoir or tape.

First, we define the joint state of a system at time t to be Z_t. The joint system (global) can be broken down into two subsystems, the interacting (local) subsystem Z_t^i, and the stationary subsystem Z_t^s. The interacting subsystem is the subsystem that is undergoing changes, while the stationary subsystem remains unchanged. This is what it means to make modular changes to a system.

Therefore, the interactions can be expressed in terms of stationary, local, and global interactions:

$$M^{\text{stationary}}_{z_t^s \to z_{t+\tau}^s} = \delta_{z_t^s, z_{t+\tau}^s} \tag{4.15}$$

$$M^{\text{local}}_{z_t^i \to z_{t+\tau}^i} = \Pr(Z_{t+\tau}^i = z_{t+\tau}^i \mid Z_t^i = z_t^i) \tag{4.16}$$

$$M^{\text{global}}_{(z_t^i, z_t^s) \to (z_{t+\tau}^i, z_{t+\tau}^s)} = M^{\text{local}}_{z_t^i \to z_{t+\tau}^i} \delta_{z_t^s, z_{t+\tau}^s} \tag{4.17}$$

Whereby the identity operator is used for the stationary interaction, such that $z_t^s = z_{t+\tau}^s$.

To derive the dissipation due to such a modular interaction, we start from the average entropy production of the universe $\langle \Sigma \rangle$ in an interval τ, which is defined as the work done in that interval, minus away the change in non-equilibrium free energy F_t^{neq}:

$$\langle \Sigma_{t \to t+\tau} \rangle = \frac{1}{T}\Big[\langle W_{t \to t+\tau} \rangle - (F_{t+\tau}^{\text{neq}} - F_t^{\text{neq}})\Big] \tag{4.18}$$

The non-equilibrium free energy is defined as the weighted sum of the joint states' internal energy U_z minus by the entropies of these joint states.

$$F_t^{\text{neq}} = \sum_z \Pr(Z_t = z)U_z - k_B T \ln 2 \; H[Z_t] \tag{4.19}$$

Taking all states to have the same internal free energy and substituting the expression for F_t^{neq} into equation 4.18, we have the generalised Landauer's Principle [53]:

$$\langle \Sigma_{t\to t+\tau} \rangle = \frac{\langle W_{t\to t+\tau} \rangle}{T} + k_B \ln 2 \; (H[Z_{t+\tau}] - H[Z_t]) \tag{4.20}$$

We should emphasise at this point that the work done $\langle W_{t\to t+\tau} \rangle$ is not the work production of the system that we have been discussing in the past chapters. Rather, this is the work required to be put into the system in order to do computation. Thus, in the context of our information ratchet, this is actually the heat $\langle Q \rangle$ absorbed from the heat reservoir.

Now, modularity will be applied. The minimum work done of a modular computation is simply the minimum amount of work required to do computation or information processing at the local subsystem, and is negative of the change in entropy of the interacting or local subsystem. Note that this is simply the generalised information processing bound as in equation 4.1.

$$\begin{aligned} \langle W_{t\to t+\tau} \rangle &\geq \langle W^{\text{local}}_{t\to t+\tau} \rangle_{\min} \\ &= k_B T \ln 2 \; (H[Z_t^i] - H[Z_{t+\tau}^i]) \end{aligned} \tag{4.21}$$

On the other hand, if we were to do computation globally without modularity, the minimum required work is negative the change in entropy of the entire system.

$$\langle W^{\text{global}}_{t\to t+\tau} \rangle_{\min} = -k_B T \ln 2 \; (H[Z_{t+\tau}] - H[Z_t]) \tag{4.22}$$

Upon substituting equations 4.21 and 4.22 into equation 4.20, we find that the minimum dissipation or entropy increase due to modular information processing or computation is:

$$T \langle \Sigma^{\text{mod}}_{t\to t+\tau} \rangle_{\min} = \langle W^{\text{local}}_{t\to t+\tau} \rangle_{\min} - \langle W^{\text{global}}_{t\to t+\tau} \rangle_{\min} \tag{4.23}$$

Thus, we can see that the modularity dissipation is simply the additional amount of work required to do a computation modularly. If we were to do the computation globally, we have $\langle W_{t\to t+\tau} \rangle = \langle W^{\text{global}}_{t\to t+\tau} \rangle_{\min}$, and there will be no modularity dissipation. Simplifying equation 4.23, we arrived at the general equation for modularity dissipation:

$$\begin{aligned} \frac{\langle \Sigma^{\text{mod}}_{t\to t+\tau} \rangle_{\min}}{k_B \ln 2} &= \frac{\langle W^{\text{local}}_{t\to t+\tau} \rangle_{\min}}{k_B T \ln 2} + H[Z_{t+\tau}^i, Z_t^s] - H[Z_t^i, Z_t^s] \\ &= H[Z_t^i] - H[Z_{t+\tau}^i] + H[Z_{t+\tau}^i, Z_t^s] \\ &\quad - H[Z_t^i, Z_t^s] + (H[Z_t^s] - H[Z_t^s]) \\ &= I[Z_t^i; Z_t^s] - I[Z_{t+\tau}^i; Z_t^s] \end{aligned} \tag{4.24}$$

Thus, the modularity dissipation is the change in mutual information between the interacting or local subsystem and the stationary subsystem, before and after the interaction of interval τ. Similar to previous section, we can better visualise this by drawing an information diagram as shown in Figure 4.6.

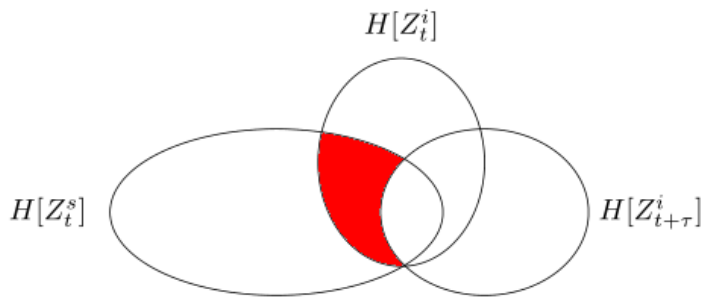


Figure 4.6: Information diagram of the interacting or local subsystem and the stationary subsystem. The red area shows the modularity dissipation. This is the mutual information lost between the local subsystem and the stationary subsystem when the local subsystem transits from $Z_t^i \to Z_{t+\tau}^i$.

In the figure, the modularity dissipation is shown as the red area. One could imagine the entropy atom $H[Z_t^i]$ moving to its new position at $H[Z_{t+\tau}^i]$ after the interval τ, and the red areas are the amount of mutual information with the stationary subsystem that were lost after the interaction interval τ. This lost of mutual information illustrates the irreversibility of modularity, and the mutual information between the interacting local subsystem and the stationary subsystem will continues to diminish over multiple intervals.

Now, let us apply this to our information ratchet system. Recall that during the ratchet's interaction transition, we have $X_N \otimes Y_N \to X_{N+1} \otimes Y'_N$, while the rest of the input and output tapes, $Y_{N+1:\infty}$ and $Y'_{0:N}$, remains unchanged or stationary. Therefore, we have:

$$\begin{aligned} Z_t^i &= (X_N, Y_N) \\ Z_{t+\tau}^i &= (X_{N+1}, Y'_N) \\ Z_t^s &= (Y_{N+1:\infty}, Y'_{0:N}) \end{aligned}$$

$$\begin{aligned} \implies \frac{\langle \Sigma_N^{\text{mod}} \rangle_{\min}}{k_B \ln 2} &= I[X_N, Y_N; Y_{N+1:\infty}, Y'_{0:N}] \\ &\quad - I[X_{N+1}, Y'_N; Y_{N+1:\infty}, Y'_{0:N}] \end{aligned} \tag{4.25}$$

4.3.1 Pattern Extractor

Unlike transient dissipation, equation 4.25 describes the ratchet's interaction with both the input and the output processes, this allow us to investigate the relation between the ratchet and the input process. To simplify our discussion, we will make some assumptions about the system. First, we will assume that the output tape is memoryless. This means that the ratchet functions as an extractor, completely extracting temporal correlations or patterns from the input tape, and outputting an uncorrelated or memoryless output process.

This means that we can remove all occurrence of Y' in the equation as the output process has no correlations with any other part of the system. Therefore, the modular dissipation of a pattern extractor is:

$$\frac{\langle \Sigma_N^{\text{ext}} \rangle_{\min}}{k_B \ln 2} = I[X_N, Y_N; Y_{N+1:\infty}] - I[X_{N+1}; Y_{N+1:\infty}] \tag{4.26}$$

This is the minimum cost of removing the patterns or temporal correlations from an input process. Like the pattern generator, the pattern extractor is not a new functionality of the ratchet. In fact, our self-synchronising ratchet in the previous chapter is exactly a pattern extractor, whereby the ratchet increases the entropy rate or 'messes up' the fully correlated period-2 pattern of 101010..., producing

work in the process.

It is straight forward to see that to minimise this, we require the first and second terms to cancel each other. This means that the joint state $X_N \otimes Y_N$ and the next ratchet state X_{N+1} must share the same amount of information with the future inputs, or in other words, is equally capable of predicting the future inputs.

For this, we requires two conditions. The first is one that we have always assumed—our ratchets are non-anticipatory, such that they cannot predict the future inputs without information from the past inputs. This gives:

$$I[Y_{N:\infty}; X_N \mid Y_{0:N}] = 0 \tag{4.27}$$

Equation 4.27 is the mutual information between X_N and $Y_{N:\infty}$, conditioned by $Y_{0:N}$. This means that there are no mutual information between X_N and $Y_{N:\infty}$ that are not already provided by $Y_{0:N}$, or one can say that all of X_N's knowledge about the future inputs $Y_{N:\infty}$ comes from the past inputs $Y_{0:N}$.

The second condition is that of a predictive ratchet. We have talked about predictive ratchets in the previous section, these are ratchets that are functions of the past inputs, such that the ratchet is able to predict the future inputs from the past inputs. This gives:

$$I[Y_{N:\infty}; Y_{0:N} \mid X_N] = 0 \tag{4.28}$$

This states that all shared information between the past and future inputs are fully captured or contained by the ratchet. While this might seems similar to the condition of non-anticipatory, they are subtly different. The condition of non-anticipatory states that the ratchet cannot get information about the future inputs from anywhere except the past inputs, whereas the condition of predictivity states that the ratchet does indeed get this information from the past inputs.

These two conditions allow us to claim that all information shared between the ratchet and the future inputs, are the same as all information shared between the past and future inputs:

$$I[X_N; Y_{N:\infty}] = I[Y_{0:N}; Y_{N:\infty}] \tag{4.29}$$

Therefore, we can write the first term of equation 4.26, as:

$$I[X_N, Y_N; Y_{N+1:\infty}] = I[Y_{0:N}, Y_N; Y_{N+1:\infty}] \tag{4.30}$$

Whereby we simply replace X_N with $Y_{0:N}$, since their mutual information with $Y_{N+1:\infty}$ is the same. Similarly for the second term, all information about the future inputs $Y_{N+1:\infty}$ that the ratchet X_{N+1} can know, are also known by its past inputs $Y_{0:N+1} = Y_{0:N}Y_N$:

$$I[X_{N+1}; Y_{N+1:\infty}] = I[Y_{0:N}, Y_N; Y_{N+1:\infty}] \tag{4.31}$$

Therefore, the first and second terms are the same and we have:

$$\begin{aligned}\frac{\langle \Sigma_N^{\text{ext}} \rangle_{\min}}{k_B \ln 2} &= I[X_N, Y_N; Y_{N+1:\infty}] - I[X_{N+1}; Y_{N+1:\infty}] \\ &= I[Y_{0:N}, Y_N; Y_{N+1:\infty}] - I[Y_{0:N}, Y_N; Y_{N+1:\infty}] \\ &= 0\end{aligned} \tag{4.32}$$

The most efficient pattern extractor is a predictive ratchet, as the modularity dissipation is zero. A predictive ratchet is one that make full use of the past inputs to predict the future inputs, it stores all information in the input's past that is relevant to the future inputs. This is exactly what happen when a ratchet is synchronised to the input's causal states. The ratchet is able to transits to the right joint states at the right input bit as it is capable of 'predicting', or 'knowing' which bit comes after. The only way for the ratchet to be able to know, is if it is a function of the past inputs, a predictive ratchet.

The principle of Requisite Complexity states that the entropy of the ratchet $H[X_N]$ must be greater or equal to the entropy of the causal states of the input $H[S_N]$. This is the direct consequence of a predictive ratchet. If the ratchet does not have enough memory to store all the information about the past inputs to predict the future inputs—i.e. $H[X_N] < H[S_N]$, we have:

$$I[Y_{N:\infty}; Y_{0:N} \mid X_N] > 0 \tag{4.33}$$

Such that there are information about the future inputs that the past inputs holds, that are not captured by the ratchet, as the ratchet does not have enough memory to hold the information.

While we can say that a predictive ratchet is the most efficient pattern extractor, what about a memoryless ratchet that simply outputs a memoryless output process like a bias coin flip, regardless of the patterns of the input process? This ratchet will be able to remove correlations or patterns from the input tape and outputs uncorrelated sequences, without having to go through the trouble of synchronising with the input process.

Being a memoryless ratchet, it cannot store any information, thus we simply remove all occurrence of X_N from our modularity dissipation, giving:

$$\begin{aligned} \langle \Sigma_\infty^{\text{ext}} \rangle_{\min} &= \lim_{N\to\infty} k_B \ln 2 \; I[Y_N; Y_{N+1:\infty}] \\ &= k_B \ln 2 \; (H - h_\mu) \end{aligned} \tag{4.34}$$

We have taken the limits of $N \to \infty$ on both sides, so that we can express the length-1 redundancy:

$$\lim_{N\to\infty} I[Y_N; Y_{N+1:\infty}] = H - h_\mu \tag{4.35}$$

Therefore, even though we can have a memoryless ratchet that always output bits 0 and 1 as a bias coin flip without any care for the inputs, the dissipation will still depends on the correlations or length-1 redundancy of the input process. Hence, we can quite safely say that a predictive ratchet is the best choice at extracting patterns or memoryful input processes.

4.3.2 Pattern Generator

We have already talked about pattern generators in the previous section, and found that retrodictive generators are the most efficient with no transient dissipation. We will see if that is also the case for modularity dissipation.

Similar to the previous section, we define a pattern generator as taking a memoryless input and turning it into a memoryful output with temporal correlations or patterns. This means that we remove all occurrence of input processes Y from modularity dissipation, as they are memoryless. This gives:

$$\frac{\langle \Sigma_N^{\text{gen}} \rangle_{\min}}{k_B \ln 2} = I[X_N; Y'_{0:N}] - I[X_{N+1}, Y'_N; Y'_{0:N}] \tag{4.36}$$

Our first hint of retrodictivity is that we want the first term to be small. The first term is the mutual information between the ratchet and the past outputs, which implies predictivity. Again, to minimise this

we require the first and second terms to cancel each other. This is again, only true with two conditions. The first is expected of a pattern generator, all correlations or mutual information between the output's past and future must be fully captured by the ratchet. This is because these correlations are generated completely by the ratchet, as there are no correlations in the memoryless input process to begin with:

$$I[Y'_{0:N}; Y'_{N:\infty} \mid X_N] = 0 \tag{4.37}$$

The second condition is of course retrodictivity, such that the ratchet states are functions of only the future outputs. Therefore, all information between the ratchet and the past outputs are fully captured by the future outputs, as these future outputs determine the ratchet states.

$$I[Y'_{0:N}; X_N \mid Y'_{N:\infty}] = 0 \tag{4.38}$$

Again, putting these two conditions together, we can replace the mutual information between the ratchet and the past outputs with the mutual information between the future outputs and past outputs, which is the first term of our modularity dissipation:

$$I[X_N; Y'_{0:N}] = I[Y'_{N:\infty}, Y'_{0:N}] \tag{4.39}$$

Similarly for the second term:

$$\begin{aligned} I[X_{N+1}, Y'_N; Y'_{0:N}] &= I[Y'_{N+1:\infty}, Y'_N; Y'_{0:N}] \\ &= I[Y'_{N:\infty}, Y'_{0:N}] \end{aligned} \tag{4.40}$$

Thus, the first and second terms are the same and cancels, giving:

$$\begin{aligned} \frac{\langle \Sigma_N^{\text{gen}} \rangle_{\min}}{k_B \ln 2} &= I[X_N; Y'_{0:N}] - I[X_{N+1}, Y'_N; Y'_{0:N}] \\ &= I[Y'_{N:\infty}, Y'_{0:N}] - I[Y'_{N:\infty}, Y'_{0:N}] \\ &= 0 \end{aligned} \tag{4.41}$$

Similar to our analysis for transient dissipation, we find that retrodictive ratchets are the best choice for pattern generation. In fact, equation 4.36 is very similar to the equation for transient dissipation in equation 4.8, with the second term being the difference, and we get the same equation if equation 4.40 is substituted into the second term of the modularity dissipation. Thus it is more general.

We have explored two different structural costs associated to information processing. The transient dissipation describes the costs of keeping excess or unnecessary information about the past, than those that are needed to predict the future. On the other hand, the modularity dissipation is the costs of doing information processing or computation modularly. This is the dissipation associated to the lost or erasure of mutual information between the interacting subsystem and the stationary subsystem. One might notice that the studies of these structural costs and dissipations are simply an effort to identify where excess information are stored or shared, and eventually erased. To minimise these costs, an effort must be made to reduce unnecessary correlations.

Modularity dissipation was suggested to be an additional costs beyond the Landauer's Principle [53]. Landauer's Principle follows from Liouville's theorem, while modularity dissipation comes from the reduction of correlations between the local or interacting subsystem, and the stationary subsystem. Designs of computational devices or systems that processes information, must take into account the modularity dissipation as an additional structural cost.

Chapter 5

Conclusion

5.1 Summary

More than a hundred years of tackling the conundrum that is Maxwell's Demon has forced us to treat information as a physical quantity, and on the same footing as the thermodynamic entropy in a thermodynamic system. Landauer's Principle tells us that the erasure of information is a dissipative and irreversible process, generating entropy in the system. This is generalised in the modified Second Law, which relates the thermodynamic entropy of a system with its informational Shannon entropy.

Modern studies focus on identifying other dissipations that arises from information processing, as well as designing an efficient Maxwell's Demon engine that functions with minimum dissipations. This brings us to the information ratchet system, this Maxwellian ratchet enables us to quantify the amount of work done and the amount of information change in the information reservoir.

However, the ratchet's implementation is a variable and affects the efficiency between the amount of work done and the amount of information change. This serves as our motivation to identify the conditions for a ratchet's structure that are required for efficient information processing, as well as their origins. By writing a program that simulates the information ratchet system, we were able to implement different ratchet designs easily, allowing us to compare between ratchet designs, as well as to quantify their efficiencies and functionalities.

Our study revealed two main results. First, there is a bound of k_BT/e that cannot be surpassed by any single ratchet design. This arises even in memoryless systems, implying that the source of the dissipation that results in this unsurpassable bound does not come from unaccounted correlations within the systems. Rather, it is a consequence of the discontinuous change in joint states during the switching transition [50].

The second result is the need for a ratchet to synchronise to the input process. This is dependent on the structure of the ratchet, and an essential condition for efficient information processing. Unlike the bound of k_BT/e, the origin of this requirement have to do with correlations within the system. This is because synchronisation requires the ratchet to be able to anticipate the next incoming bit, thus mutual information between the two is a must. This is further hinted by the principle of Requisite Complexity. Two different ratchets performs almost the same as they have the same amount of entropy or memory, and it must be greater than that of the input process.

The origins of these two results must have come from the costs and dissipations of information processing, as those are the fundamental principles of information processing, and the efficiencies and performance of our ratchets are simply the consequences of them. Exploring two structural costs of information processing, we are able to derive the types of ratchets that minimise the costs and dissipations, and it is found that the condition of synchronisation and the principle of Requisite Complexity comes from minimising the modularity dissipation.

Finally, we can quite surely say that there are indeed structural costs to information processing devices that are not captured by Landauer's Principle. These are the various dissipations that we have

explored, as well as the existence of an unsurpassable bound of $k_B T/e$. The Landauer's Principle states that the erasure of information must have an associated cost or dissipation, which is captured in the modified Second Law, and its extension the IPSL. However, it does not describe the implementation of the erasure of information, which have structural costs and dissipations such as the transient dissipation and the modularity dissipation.

There are also indeed best choices of structural designs for a given information content. In the context of a ratchet's design, this is the need to design the ratchet to be able to synchronise to the input processes. However, in the context of more fundamental information-theoretic arguments, we find that the best choice of structural design is simply the design that minimises the various structural costs of transient and modularity dissipations.

Therefore, we can conclude that efficient information processing must depend on the various structural costs and dissipations of the implementation of the information processing device, which are not captured by the Landauer's Principle. An efficient device must minimise these costs, and this is partly done so by careful control of the amount of correlations or mutual information accumulated in the system. This is demonstrated in transient dissipation as dissipations come from the ratchet keeping useless mutual information between itself and the past, and also in modularity dissipation as the erasure of mutual information between subsystems.

There are of course still various other structural costs that were not investigated, such as the origin of the bound of $k_B T/e$ which does not come from the dissipation of mutual information, and further studies can hope to uncover these unaccounted structural costs. Understanding of these structural costs can enable better designs of computing devices, that works more efficiently with little dissipation. Insights into the functioning of biological enzymes that utilises Maxwellian work can also be gained, allowing designs of Maxwell's Demon enzyme engines that can fulfils specific tasks that are medical in nature. As mentioned, the information reservoir can have symbols of $\mathcal{Y} \in \{A, T, G, C\}$ for DNA bases, allowing the study of RNA transcoding and correcting errors from mutations, similar to a ratchet working with noisy inputs. Overall, the study of information processing which is how information is created, stored, transferred, shared, erased, and how they are connected to the entropy dissipation in a thermodynamic system, can work towards a physical theory of information.

5.2 Future Works

5.2.1 Generalised Systems

Further studies of the efficiencies and costs of information processing can be approached from a more general perspective, instead of being specific to the information ratchet system. For example, the information ratchet system can be expressed as a multipartite framework, and ratchet-tape dynamics can be seen as interactions between causal networks, whereby the nodes of the networks are the system observables, and the edges are the relationship between them.

A hint of this is that the modularity dissipation that was derived in equation 4.24 is similar to another form of Second Law that is more widely used in information flows in causal networks [23]. This Second Law comes from the generalised Integral Fluctuation Theorem (IFT):

$$\begin{aligned} \langle \exp[-\Delta S + \Delta I] \rangle &= 1 \\ \implies \langle \Delta S \rangle &\geq \langle \Delta I \rangle \\ \langle \Delta S \rangle_{\min} &= \langle \Delta I \rangle \end{aligned} \tag{5.1}$$

Whereby ΔS is the change in entropy of the whole system, and ΔI is the change in mutual information of two interacting systems. The study of causal networks can account for dissipations from information flows between the ratchet and the tape. The measure of information flows can be described by the transfer entropy [57], which were missing in our study. For example, another proposed Second Law bound also derived from IFT in a bipartite system takes into account the transfer entropy [24]:

$$\langle \Delta S_X + T_{X \to Y} \rangle \geq \langle \Delta S_X \rangle + \langle h_{\mu_X} \rangle \tag{5.2}$$

Whereby ΔS_X is the change in entropy of system X, h_{μ_X} is the entropy rate of system X, and $T_{X \to Y}$ is the transfer entropy from system X to system Y. However, despite more than a decade long use of transfer entropies, there have been recent criticisms of it, suggesting that transfer entropies can overestimate information flow or underestimate correlations [58]. Other measures of information flows have also been proposed instead of the transfer entropy [59].

5.2.2 Multiple Ratchets and Complex System Designs

On the other hand, we can still study the information ratchet system by constructing more elaborate ratchet and tape systems. Not only does our program allows for fast prototyping of systems with different ratchet's transition matrices M, and input tape processes, our program also allow us to construct and engineer various ratchets and tapes structures. One could imagine an input tape that have periodic boundary conditions, which joins itself at the ends, forming a loop. This will have implications on the derivations for transient dissipation as now there are correlations or mutual informations between $Y'_{0:N}$ and $Y_{N:\infty}$, which we assumed there were none in equation 4.7.

Another interesting design we can have is a system with multiple ratchets processing a single input tape process. There will now be implications to modularity dissipation as there are now multiple interacting subsystems. One could also imagine this as having the second ratchet's input process being the output process of the first ratchet. This is shown in Figure 5.1.

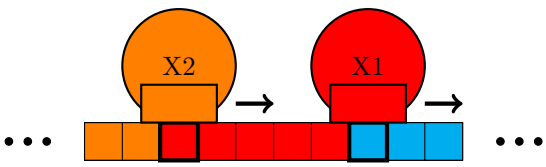


Figure 5.1: Two ratchets system. Ratchet X1's output process serves as ratchet X2's input process.

Recall in Chapter 3, we see in Figure 3.19 that the change in information content is not maximum after passing the input process through the most optimal ratchet, and that the maximum work production cannot surpass $k_B T/e$. If instead we pass the output process to a similar second ratchet like in Figure 5.1, we find that the maximum work production can surpass $k_B T/e$ shown in Figure 5.2.

Therefore, we see that having multiple ratchets working together can improve the functionality and efficiency of information processing. One can also design multiple memoryless ratchets that can work together to leverage memoryful input processes, a feat that cannot be done by a single memoryless ratchet.

More complex systems can also be built. A system described in [56] have a pattern extractor extracting patterns, and thus producing work. This produced work is then used to generate the same pattern, and the system runs cyclically. This is shown in Figure 5.3. Given the most efficient ratchets with minimum dissipation of a predictive extractor and a retrodictive generator, the cycle should be able to be ran without any losses.

Not only could the designs of complex ratchet and tape structures identify additional dissipations that are not apparent in single ratchet systems, they could also provide the same functionalities of single ratchet systems, while minimising dissipations as multiple ratchets work together. The efficiencies of the Maxwell's Demon engine could also be increased by having multiple ratchets.

Overall, while our study focus on the conditions for efficient information processing, as well as to minimise the structural costs of the ratchet's implementation, they are subsets of the more general problem of

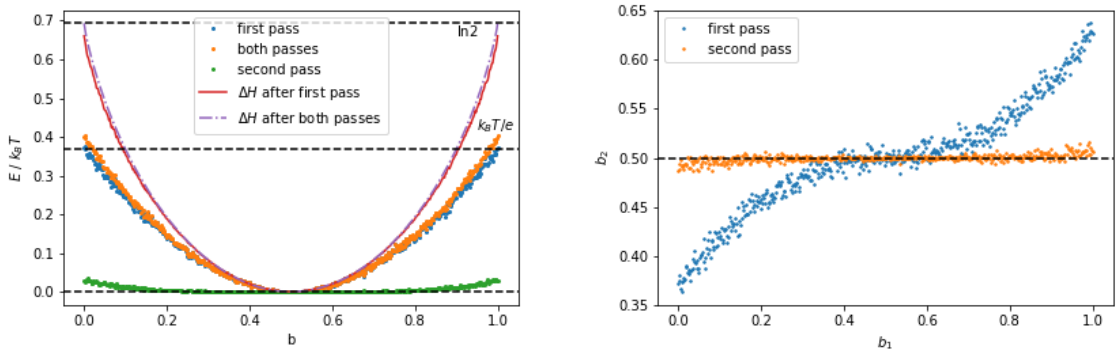


Figure 5.2: Left – Plots of maximum work production rate $\langle W \rangle_{\max}$, and its corresponding $\ln 2\ \Delta H$. The work production rate cannot surpass k_BT/e after being processed by a ratchet once. A second pass by another similar ratchet can surpass k_BT/e (orange plot). The change in information content also reach the Landauer bound of $\ln 2$. Right – The bias values of the input process. For initial bias values 0 and 1, the output bias value did not reach the maximum entropy value of 0.5. After a second pass by another similar ratchet, the bias values reached 0.5, indicating that all information in the tape was leveraged.

identifying the dissipations or entropy productions of information processing, for which we have provided two examples in equations 5.1 and 5.2. By generalising the information ratchet system, or by designing clever and elaborate ratchet and tape structures, it might be possible to take into account of other forms of structural costs or dissipations including the ones from information flows.

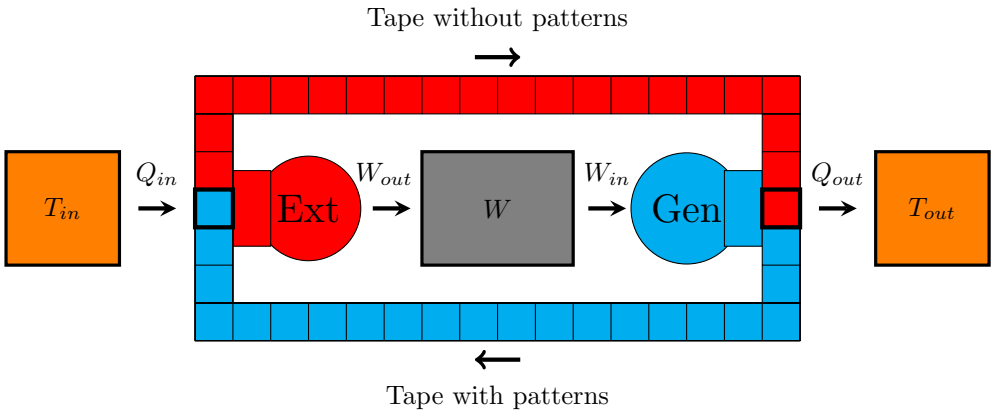


Figure 5.3: A pattern extracting and generating cycle described in [56]. An extractor will extract patterns from the memoryful tape (cyan), outputting a patternless or memoryless tape and producing positive work. Another generator will then generate patterns from the memoryless tape (red), consuming the work from the work reservoir. The cycle will continues to run if the ratchets are efficient with no dissipation during the pattern extracting and generating processes.

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